UNIVERSITY OF SANTO TOMAS
COLLEGE OF INFORMATION AND COMPUTING SCIENCES
Department of Information Systems
2nd Semester AY 2025 - 2026

Demand Forecasting and Prescriptive Inventory Management Dashboard for the USTore

IS 26312 (Capstone Project 1)

Prepared By :

Marco C. Arante

Gabriel Paul Auren Z. Gambe

Neil Sam R. Perez

Tyrone Gryneth S. Yazon

A.P. Princess Arleen Zamora

May 2026

Table of Contents

Chapter 1: Introduction 1
1.1 Company Background 2
1.2 Project Context (Business Analytics) 4
1.3 Purpose and Objectives 9
1.3.1 General Objectives 9
1.3.2 Specific Objectives 10
1.4 Scope and Project Constraints 12
1.4.1 Project Scope 12
1.4.2 Project Constraints 14
1.5 Conceptual Framework of Data Integration 17
1.6 Project Milestones/Timeline 21
Chapter 2: Review of Related Literature, Studies, Systems, and Technologies 24
2.1 Review of Related Literature and Studies 24
2.1.1 Inventory Management and Demand Forecasting in Retail 24
2.1.2 Manual Record-Keeping and Its Operational Consequences 27
2.1.3 FSN Analysis and Inventory Classification 28
2.1.4 Time-Series Forecasting Methods in Retail Analytics 29
2.1.5 Dashboard-Based Decision Support in Retail and Institutional Operations 33
2.1.6 Synthesis 34
2.2 Review of Related Systems and Technologies 37
2.2.1 Existing Inventory Management and POS Systems 37
2.2.2 Non-Transactional Digital Tallying and Lightweight Data Capture Systems 39
2.2.3 Python-Based Analytics Frameworks for Inventory and Demand Forecasting 40
2.2.4 Power BI Dashboard Implementations in Inventory and Retail Contexts 41
2.2.5 Synthesis 42
2.3 Chapter Synthesis 44
Chapter 3 : Methodology 46
3.1 Data Gathering (with ETL) 49
3.1.1 Data Sources 50
3.1.2 Data Extraction 57
3.1.3 Data Transformation 59
3.1.4 Data Loading 62

3.2 Presentation of Star Schema 63
3.3 Business Analytics Model and Testing 68
3.3.1 Fast, Slow, and Non-moving (FSN) Analysis Model 68
3.3.2 Demand Forecasting Model 71
3.3.3 Reorder Point, Safety Stock, and Economic Order Quantity Calculation 75
3.3.4 Model Validation and Testing Protocol 79
3.4 Business Analytics Tools, Techniques, and Specific Applications 81
3.5 Risk Assessment and Mitigation 87
Bibliography 94

Chapter 1: Introduction

Across global retail markets, the management of inventory through data-driven forecasting has become a core operational differentiator (Famoti et al., 2025). Supply chain research consistently identifies organizations that rely on assumption-based replenishment practices as significantly more susceptible to stockouts, overstocking, and cascading supply chain disruptions that erode long-term profitability. The growing adoption of digital tallying, automated point-of-sale systems, and predictive analytics platforms reflects an industry-wide recognition that clean, real-time transaction data is the foundation on which all meaningful demand forecasting must rest (Prabu, 2023).

Within the Philippine retail landscape, the transition toward digitized inventory management remains particularly challenging for small-to-medium enterprises and institutional retail operations (Habaradas et al., 2024). According to the Department of Trade and Industry, a substantial portion of local retail establishments continue to depend heavily on manual record-keeping, a practice that consistently produces incomplete data baselines and limits a business's capacity to anticipate future demand (Hua, 2023). University-based and campus retail stores operate under an additional layer of constraint, as strict institutional financial regulations and the absence of enterprise-grade technical infrastructure frequently prevent the adoption of standard point-of-sale solutions that commercial establishments take for granted (Laudon et al., 2022).

1.1 Company Background

The University of Santo Tomas Store (USTore) is the official merchandise outlet of the University of Santo Tomas, situated within the España campus in Manila. The USTore serves a diverse clientele composed of enrolled students, alumni, faculty, administrative staff, and campus visitors, offering a catalog that includes institutional apparel, memorabilia, and university-branded goods. Operationally, the USTore functions under a consignment business model. Under this arrangement, the university does not purchase inventory upfront; instead, participating suppliers deliver merchandise directly to the store, and the university compensates each supplier on a monthly basis strictly according to the volume of items confirmed as sold within that billing period. The execution of these monthly payments requires coordination across two university offices, that is, the UST Purchasing Office, which is responsible for generating Purchase Orders, and the Finance Department, which issues the corresponding payment checks to each consignment supplier based on batch sales reports produced by the store.

The USTore's core operational challenge originates in a regulatory constraint that distinguishes it from commercial retail environments. University policy, in alignment with Bureau of Internal Revenue regulations governing third-party financial systems, prohibits the USTore from deploying automated Point-of-Sale (POS) systems for processing customer transactions. As a result, the store's staff record sales manually using paper tally sheets that are categorized per consignment supplier. While this manual process satisfies the regulatory requirement, it introduces substantial operational risk during high-traffic periods such as university enrollment events,

foundation days, and varsity competitions. During these peak periods, the volume of simultaneous transactions frequently exceeds the capacity of the manual tally system, resulting in missed or unrecorded sales entries.

The consequence of these missed tallies is not only about bookkeeping inconvenience. Because the USTore's inventory baseline is compiled from accumulated daily tally data, any unrecorded transaction directly distorts the store's understanding of its actual stock levels. At the end of each billing cycle, management addresses this discrepancy through a secondary manual count process, identifying sold items that were not tallied and rolling the corresponding payment into the supplier's following month's settlement. While this mechanism ensures that suppliers are eventually compensated, the perpetual rollover effect means that the store's real-time inventory record is never fully accurate. A store operating on a consistently distorted inventory baseline cannot calculate reliable stock depletion rates, cannot distinguish fast-moving items from slow-moving ones, and cannot anticipate when critical merchandise will fall below a safe stock threshold. The practical outcome is a store management team that relies on experience-based guesswork rather than data-backed judgment; a condition that results in recurrent and unanticipated stockouts of popular merchandise, lost revenue opportunities, and compounded administrative burden at the close of every billing month.

To resolve the identified operational bottlenecks, the USTore requires a transition from its current error-prone manual tallying system to a structured, digitized approach to transaction recording and inventory tracking. The proposed improvement centers on the development of a Demand Forecasting and Prescriptive Inventory

Management Dashboard; a Business Analytics system designed to capture clean, itemized daily sales data, classify inventory by movement velocity, generate demand forecasts calibrated to the academic calendar, and produce prescriptive restocking advisories for the store's management. The dashboard will additionally automate the generation of accurate monthly batch sales reports, directly addressing the pain point experienced by the UST Purchasing Office and the Finance Department in executing timely supplier compensation.

1.2 Project Context (Business Analytics)

The proposed Demand Forecasting and Prescriptive Inventory Management Dashboard is designed as a Business Analytics capstone project, not a software engineering deliverable. The distinction is deliberate: the primary output of the project is a decision-support dashboard driven by statistical forecasting models, rather than a transactional application. The system will be constructed in four functional layers that collectively transform raw, manually entered daily sales data into prescriptive restocking intelligence.

The first layer is the Data Capture Layer. A lightweight digital tallying interface will be developed to replace the current paper-based tally sheets. The interface will allow USTore staff to log individual transactions by item, quantity, date, and assigned consignment supplier. Critically, the interface will be designed to function exclusively as an internal inventory counting tool; it will not process monetary transactions, calculate customer-facing totals, or generate Bureau of Internal Revenue receipts, thereby

preserving full compliance with university policy and BIR regulations. The intended technology for the tally interface is a simple web-based form; the specific front-end framework and database back end are evaluated in Section 2.2.2 and formalized in Chapter 3. The interface will also include a dynamic holiday mapping feature that allows USTore staff to tag specific dates as store closures or suspension days directly through the UI, ensuring that these dates are recorded as store closures rather than as ordinary zero-sales days.

The second layer is the Data Processing Layer. Historical transaction records held by the USTore, spanning physical inventory count sheets and manual tally sheets from 2023 to the present, will be ingested alongside the new clean data produced by the tallying interface. A Python-based Extract, Transform, and Load (ETL) pipeline developed using the pandas library will clean and standardize the historical data, resolve the price-grouping discrepancies present in pre-digitization records, and produce an itemized, date-stamped sales baseline. The underlying data model will follow a Star Schema architecture, with a central Fact_Sales table surrounded by dimension tables for products and dates. The Star Schema structure is selected because it optimizes analytical query performance and aligns with the dimensional modeling conventions required by the project's visualization layer (Adamson, 2006; Morfonios et al., 2009).

The third layer is the Analytics Layer. The Python-based analytical engine will execute two sequential processes. First, Fast-Slow-Non-moving (FSN) Analysis will classify each SKU in the USTore catalog according to its historical sales velocity, producing three inventory segments: fast-moving items that require proactive

restocking, slow-moving items that may be candidates for promotional intervention, and non-moving items that represent potential dead stock (Tambunan et al., 2018). Within the fast-moving segment, items that exhibit high sales velocity but are constrained by limited consignment stock or short selling windows are further designated as High-Velocity Limited (HVL), as detailed in Section 3.3.1. This classification directly informs the second process: Time-Series Demand Forecasting. The project will forecast 30-day demand for each product category and for each SKU classified as fast-moving (including High-Velocity Limited items), which represent the items where forecast accuracy carries the highest operational consequence. Category demand is forecast with a six-month (180-day) moving average; each fast-moving SKU is forecast with an equal blend of (a) its share of its category's sales over the last 30 days applied to the category level and (b) the Teunter-Syntetos-Babai (TSB) method for intermittent demand. These methods were selected by comparing candidate methods, including Prophet, on the USTore's own sales data (Sections 3.3.2 and 3.3.4). The academic calendar enters the forecast through a calendar adjustment: using the is_enrollment_period, is_exam_week and is_sem_break flags in Dim_Date, each 30-day forecast is lowered when the coming 30 days hold a quieter mix of days (semestral breaks, examination weeks) than the period it was learned from; it is never raised. Because the USTore's sales volumes are influenced by the academic calendar, including enrollment periods, examination weeks, university events and semestral breaks (Shapiro et al., 2026), candidate methods (Prophet, ARIMA, exponential smoothing, intermittent-demand methods such as Croston, SBA and TSB, and moving averages) were compared on the USTore's own data. The selected model is a

six-month moving average for each product category and, for each fast-moving SKU, an equal blend of the item's category share and the TSB method, with an academic-calendar adjustment that lowers the forecast when semestral breaks or examination weeks are ahead. TSB is used for items because most item-days record no sales and its forecast decays when an item stops selling. The forecasting model will target a Mean Absolute Percentage Error (MAPE) of 20 percent or lower as the primary acceptance criterion during validation, supplemented by Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) as diagnostic metrics, all compared with a naive forecast that repeats the previous period. The result is reported whether or not the target is met, and the reorder recommendations are judged separately against the USTore acceptance standard (Section 3.3.4). The Reorder Points, Safety Stock thresholds, and Economic Order Quantities (EOQ), calculated from each item's actual sales over the last 365 days and presented alongside these demand forecasts, constitute the prescriptive component of the system. The prescriptive layer employs an EOQ-based optimization model to minimize the total inventory cost composed of ordering costs incurred per replenishment cycle and holding costs incurred per unit stored over time. This optimization is subject to a cycle service level constraint which, in operational terms, represents the target probability that the store will not run out of stock while waiting for a supplier delivery during the lead time window. The EOQ formula determines the optimal number of units to order per replenishment cycle, while the ROP determines the stock level at which that order must be placed. Together, these components ensure that the dashboard does not merely predict future demand

but prescribes specific restocking actions, optimal quantities, and timing for each qualifying SKU.

It should be noted that the USTore's customer base extends beyond enrolled students to include alumni, faculty, administrative staff, and campus visitors. These non-student segments may exhibit purchasing patterns that do not align with the academic calendar adjustment used in the forecasting model. For example, alumni purchases may spike during homecoming events or graduation ceremonies rather than during enrollment periods. While the current model does not segment demand by customer type because the USTore's tally records do not capture buyer identity, the system partially mitigates this limitation through the Event_Log feature. By explicitly flagging events that historically draw large non-student crowds, such as alumni homecoming events, baccalaureate masses, and graduation ceremonies, the Event_Log records these demand spikes without requiring individualized customer data; using these flags as a forecast input is moved to the Recommendations until enough events have been logged. As the Digital Tallying Interface accumulates data over successive semesters, future iterations of the model may explore additional calendar or event adjustments to account for more granular non-student demand drivers if the data supports it.

The fourth layer is the Visualization and Advisory Layer. Analytical outputs from the Python engine will be integrated into a Microsoft Power BI dashboard that serves as the primary decision-support interface for USTore management. Power BI is

selected over Tableau for this deployment because of its compatibility with the Microsoft Office ecosystem prevalent in Philippine university administrative environments, its free desktop-tier license that requires no institutional procurement, and its documented capability to generate exportable PDF reports; which is a functional requirement driven by the Finance Department's need for automated monthly batch sales documents (Garudasu et al., 2025). The dashboard will provide five functional views: a stock status view with color-coded safety threshold alerts, an FSN category classification view, a demand forecast view showing projected depletion rates, an automated batch sales report generation module, and calendar-contextual interpretation cards that surface forward-looking narrative guidance tied to upcoming demand-influencing academic calendar events.

The technological architecture described above integrates two primary functional areas. Functional Area 1 encompasses Sales and Inventory Operations, covering the digital capture of daily transactions through the tallying interface, real-time stock depletion tracking, and FSN item categorization. Functional Area 2 encompasses Procurement and Supplier Management, covering the generation of data-backed restocking advisories, the automated production of monthly batch sales reports, and the delivery of replenishment directions to consignment suppliers. The cross-functional integration of these two areas is the core value proposition of the proposed system, as it transforms isolated, error-prone sales data into actionable procurement intelligence (Laudon et al., 2022).

1.3 Purpose and Objectives

1.3.1 General Objectives

The project aims to develop a Demand Forecasting and Prescriptive Inventory Management Dashboard for the USTore that establishes a clean, digitized foundation for daily sales data capture and leverages time-series forecasting to generate reliable restocking advisories, thereby transitioning the store from a reactive, assumption-based replenishment decision-making approach wherein restocking quantities are determined through experiential judgment rather than quantified demand projections, to a proactive, data-backed operational model driven by time-series forecasting, prescriptive Reorder Point calculations, and Economic Order Quantity optimization that determines not only when to reorder but also how much to order for each qualifying SKU.

1.3.2 Specific Objectives

1. To construct a Digital Tallying Interface that records daily itemized USTore sales transactions by item, quantity, date, and consignment supplier, functioning strictly as an internal inventory counting tool in compliance with university policy and BIR regulations, with an initial deployment target of the first month of the Capstone 2 implementation semester, thereby producing the clean, foundational data required to drive the prescriptive dashboard.

2. To perform Fast-Slow-Non-moving (FSN) Analysis on the USTore's historical and digitally captured sales records in order to classify inventory items by movement velocity into three distinct categories: fast-moving, slow-moving, and non-moving; providing the categorization baseline that informs subsequent forecasting and restocking threshold calculations. The model will utilize the 80th percentile as the primary cutoff, and classification validity will be assessed through threshold sensitivity analysis conducted at the 75th and 85th percentiles. Borderline items will be flagged for cross-referencing against USTore management's experiential knowledge of item demand behavior, thereby establishing the category-specific decision rules required for the prescriptive dashboard.

3. To formulate a predictive analysis of the USTore's future demand by forecasting the next 30 days of sales for each product category with a six-month moving average, and for each SKU classified as fast-moving and High-Velocity Limited through FSN Analysis with an equal blend of category share and the Teunter-Syntetos-Babai (TSB) intermittent-demand method, both adjusted for the academic calendar and selected by comparing candidate models including Prophet, targeting a Mean Absolute Percentage Error (MAPE) of 20 percent or lower as the primary acceptance criterion during model validation for standard academic semester periods. MAE and RMSE are reported for every period, including semestral breaks, against a naive baseline, with results reported per 30-day test window to reflect the UST academic calendar's seasonal demand patterns.

4. To calculate the Reorder Point (ROP), Safety Stock level, and Economic Order Quantity (EOQ) for each fast-moving and slow-moving inventory item with sales in the last 365 days, utilizing each item's actual sales over that period, a safety stock sized from the item's own measured shortfalls, lead time estimates provided by the USTore and its consignment suppliers, and estimated ordering and holding cost parameters. Items with no sales in the last 365 days are flagged for manual review instead of being given a reorder point. The Safety Stock will be sized from each item's own measured shortfalls, at a service level set by the item's service tier (servable, partial or not stockable) rather than by a default z-score per FSN class (Section 3.3.3). The EOQ will determine the optimal reorder quantity that minimizes the total of ordering and holding costs per replenishment cycle. All values will be expressed in units of physical stock count and validated during Capstone 2 against beginning inventory counts and supplier lead time estimates to confirm their operational accuracy, establishing the specific optimal reorder triggers that will be surfaced through the prescriptive dashboard.

5. To design a Microsoft Power BI dashboard that integrates all upstream analytical outputs into a unified prescriptive decision-support interface for USTore management. The dashboard will visualize the USTore's estimated stock status, FSN item classifications, 30-day demand forecasts, Reorder Point alerts, and EOQ-derived optimal order quantities. It will incorporate color-coded safety stock alert triggers, category-specific decision rules for fast-moving, slow-moving, and non-moving items, and an automated

monthly batch sales report export function to support timely supplier compensation by the UST Purchasing Office and Finance Department.

1.4 Scope and Project Constraints

1.4.1 Project Scope

The scope of the project encompasses the design, development, and initial testing of the Demand Forecasting and Prescriptive Inventory Management Dashboard for the USTore. The data scope covers two source types: historical operational records held by the USTore, including beginning monthly physical inventory count sheets and manually produced sales tally records spanning from 2023 to 2026, and prospective clean transaction records generated through the proposed Digital Tallying Interface beginning from the point of system deployment. All financial computations and cost-related metrics produced by the dashboard will be expressed in Philippine Pesos (PHP).

The analytical scope encompasses three sequential processes. First, an FSN Analysis module will classify all active USTore SKUs by sales velocity using the full available historical and prospective dataset. Second, a demand forecasting module will generate 30-day demand projections for each product category, using a six-month moving average, and for each SKU classified as fast-moving and High-Velocity Limited through FSN Analysis, using an equal blend of category share and the Teunter-Syntetos-Babai (TSB) method, both adjusted for the University of Santo Tomas academic calendar to account for seasonal demand fluctuations during

enrollment periods, examination weeks, and semestral breaks. Third, a Reorder Point, Safety Stock, and Economic Order Quantity (EOQ) calculation module will produce item-level restocking thresholds and optimal reorder quantities based on each item's actual sales over the last 365 days, supplier lead times, and estimated ordering and holding cost parameters.

The system output scope encompasses four deliverables accessible through the Power BI dashboard: a real-time estimated stock status view with safety threshold alerts, an FSN-categorized product inventory view, a demand forecast view showing projected depletion rates per fast-moving SKU, and an automated monthly batch sales report exportable as a PDF document for submission to the UST Purchasing Office and Finance Department. The underlying data model will follow a Star Schema dimensional structure, formally presented in Chapter 3 of the manuscript.

1.4.2 Project Constraints

The proposed system is subject to the following organizational and technical constraints. First, in compliance with university policy and BIR regulations, the Digital Tallying Interface is strictly constrained to function as an internal inventory counting tool. The interface will not process monetary customer transactions, calculate customer-facing payment totals, or produce official BIR-compliant sales receipts. The automated data workflow within the proposed system from tallying through ETL processing to dashboard visualization constitutes the system's automation scope; physical supply chain activities, including the generation of Purchase Orders and the

issuance of supplier payment checks, remain the administrative responsibility of the UST Purchasing Office and the Finance Department respectively.

Second, the historical data available for model training spans 2023 to 2026 and is subject to quality limitations. Records predating the Digital Tallying Interface implementation were produced under the manual price-grouped tally system and carry inherent inaccuracies at the item level. Because the proportional imputation applied to these price-grouped historical records introduces estimation uncertainty at the item level, imputed records will be weighted at 0.5 in the FSN classification to reduce their influence. The forecasting model will treat pre-digitization records as a baseline estimate; forecast reliability is expected to improve progressively as the volume of clean, digitally captured data accumulates over successive academic semesters.

Third, the project does not include the automated execution of procurement transactions. The dashboard will generate prescriptive restocking advisories and exportable batch sales reports that inform purchasing decisions, but the act of placing orders with suppliers and issuing payment is outside the system's functional boundary.

Fourth, the data scope is limited to USTore consignment merchandise transactions only. Sales data from other university commercial establishments are not within the project's coverage.

Fifth, model performance during semestral break periods may be limited by data sparsity, as transaction volumes during those intervals are significantly lower than during the academic semester. Due to this data sparsity, forecast reliability during semestral break periods is expected to be lower, and the forecasting model's

performance will be evaluated separately for sparse and dense data periods in Chapter 4.

Sixth, the supplier lead time used in the Reorder Point (ROP) calculation is a generalized verbal estimate from USTore management rather than a measured value derived from transactional data. Because the system lacks historical data on exact delivery windows, this reliance on an estimated variable introduces an unavoidable degree of uncertainty into the prescriptive threshold calculations.

Seventh, the FSN classification model's primary velocity metric, Average Daily Units Sold, is calculated from episodic tally dates for historical records rather than continuous daily observations. This means items tallied on only a few dates per month will have their velocity estimated from a small number of data points, introducing classification uncertainty for items near the threshold boundary. This limitation is mitigated through the threshold sensitivity analysis conducted at the 75th, 80th, and 85th percentile cutoffs documented in Chapter 3.

Eighth, the Safety Stock formula's standard deviation of daily demand is derived from the same sparse and imputed historical dataset. A standard deviation calculated from episodic or imputed records is a less precise estimate of true demand variability than one derived from continuous daily observations. The reliability of Safety Stock calculations is therefore expected to improve progressively as the Digital Tallying Interface accumulates clean daily records over successive academic semesters.

Ninth, the ordering cost and holding cost parameters used in the Economic Order Quantity (EOQ) calculation are generalized estimates provided by USTore

management and the UST Purchasing Office rather than values derived from detailed cost accounting records. The USTore does not currently track the per-cycle administrative cost of placing a restocking request or the per-unit cost of storing inventory over time. Because EOQ output is sensitive to these cost inputs, the system will include a sensitivity analysis that computes total inventory cost at the EOQ value and at alternative order quantities to confirm that the EOQ recommendation is robust to reasonable variation in the cost estimates. This sensitivity analysis will be reported per SKU in Chapter 4.

Tenth, Events not published in the official UST collegiate calendar such as college days, sports fests, and token seasons are captured through a staff-operated event flagging feature in the Digital Tallying Interface. Until sufficient flagged event records accumulate over multiple academic cycles, the forecasting model's ability to forecast demand for these unscheduled event periods may be limited and forecast reliability for those specific dates will be reported separately in Chapter 4.

1.5 Conceptual Framework of Data Integration

The conceptual framework presented in Figure 1 illustrates the general flow of data for the proposed system. It maps out how the USTore's manual daily sales records will be collected, processed, and transformed into useful inventory insights. Ultimately, this diagram highlights how the system connects two critical functional areas: Sales and Inventory Operations, and Procurement and Supplier Management.

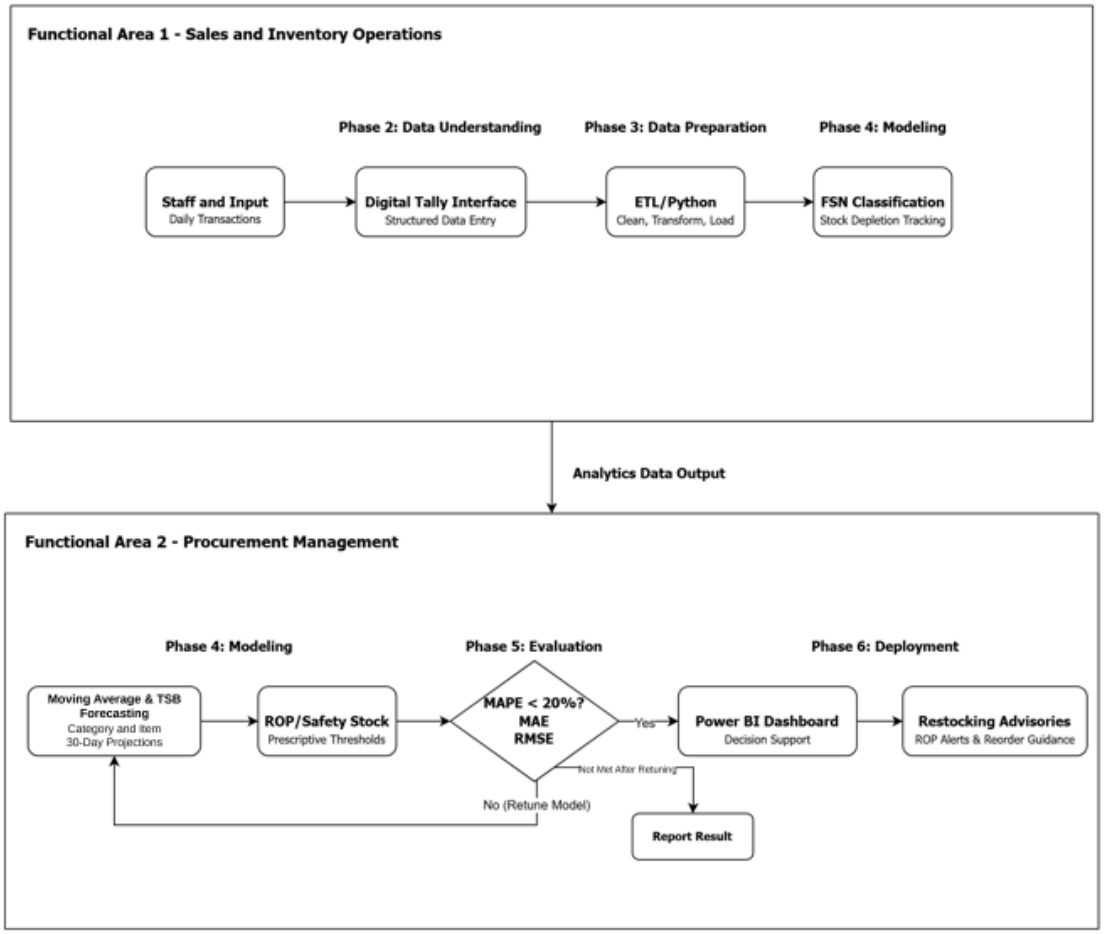


Figure 1. *Conceptual Framework of Data Integration*

The conceptual framework presented in Figure 1 illustrates the system architecture through which data integration transforms the USTore's fragmented, manual transaction records into actionable supply chain intelligence. The framework is organized across two functional areas, each mapped to the corresponding phases of the CRISP-DM methodology, reflecting the project's iterative analytical design.

The first functional area covers Sales and Inventory Operations and encompasses three sequential phases. In Phase 2, Data Understanding, USTore staff

input daily transactions through the Digital Tallying Interface, which replaces paper-based tally sheets with a structured digital data entry tool. Clean, itemized daily sales data captured through the interface, combined with the available historical inventory count records spanning 2023 to 2026, constitutes the raw input of the entire system. The integrity of every subsequent analytical output is directly dependent on the quality of data produced at this stage. In Phase 3, Data Preparation, a Python-based ETL pipeline cleanses the combined historical and prospective dataset, resolves item-level discrepancies introduced by the historical price-grouped tally format, and structures the validated data into a Star Schema dimensional model. The Star Schema serves as the data warehouse foundation that enables efficient querying for both analytical computation and dashboard reporting. In Phase 4, Modeling, the structured data is subjected to FSN Classification, which segments inventory by movement velocity and identifies items warranting proactive management, and produces stock depletion tracking outputs that serve as the analytical data output passed to Functional Area 2.

The second functional area covers Procurement and Supplier Management and begins where the first functional area concludes. Phase 4 continues with Moving Average and TSB Forecasting, which produces 30-day demand projections for each product category and each fast-moving item, adjusted for the academic calendar, and the Reorder Point, Safety Stock, and Economic Order Quantity calculation module, which computes prescriptive restocking thresholds and optimal reorder quantities from each item's actual sales over the last 365 days. In Phase 5, Evaluation, the forecasting models (the category moving average and the item category-share and TSB blend) are

assessed against a Mean Absolute Percentage Error of 20 percent or lower as the primary acceptance threshold, with Mean Absolute Error and Root Mean Square Error reported as secondary metrics and compared against a naive baseline forecast. Models that fail to meet the threshold are returned to Phase 4 for retuning and comparison with alternative methods; if none meets it, the result is reported and the forecasts do not drive restocking. The reorder recommendations are judged separately against the USTore acceptance standard (Section 3.3.4). In Phase 6, Deployment, validated model outputs are integrated into a Microsoft Power BI dashboard that serves as the primary decision-support interface, delivering restocking advisories with Reorder Point alerts and reorder guidance directly to USTore management. The cross-functional nature of this final phase represents the system's primary value proposition: the transformation of isolated, error-prone sales events into timely, data-backed procurement decisions.

The five functional views of the Power BI dashboard are sequenced to guide USTore management through a coherent decision-making narrative that mirrors the data flow illustrated in Figure 1. The Stock Status View serves as the entry point, surfacing color-coded inventory alerts derived from the Data Capture and Processing layers that immediately communicate which items require attention. From there, the FSN Classification View provides the analytical context behind each alert, explaining whether a flagged item is fast-moving, slow-moving, or non-moving, and therefore what type of management response is appropriate. The Demand Forecast View then extends this understanding forward in time, presenting 30-day demand forecasts annotated with upcoming academic calendar events to answer how much stock will be

needed and when. The Restocking Advisory View presents specific prescribed actions, including the reorder point, safety stock level, and recommended order quantity calculated from each item's actual sales over the last 365 days, alongside plain-language calendar-contextual guidance tied to upcoming demand-influencing events. Finally, the Automated Batch Sales Report closes the narrative loop by converting the same clean transaction data into an exportable PDF document for supplier compensation, directly serving the UST Purchasing Office and Finance Department. A user who moves through these five views in sequence will have progressed from awareness of a stock problem, through its cause and projected trajectory, to a specific prescribed action, entirely within a single unified interface.

1.6 Project Milestones/Timeline

The development of the Demand Forecasting and Prescriptive Inventory Management Dashboard for the USTore spans two consecutive terms: IS 26312 (Capstone Project 1), covering project initiation, data gathering, and full documentation of Chapters 1 through 3; and IS 26316 (Capstone Project 2), covering system implementation, results analysis, and final defense. The following table presents the specific activities and their corresponding timeframes across both terms.

IS 26312 – Capstone Project 1

Group 8 | Section 3ISA

ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
Client Finding and Group Emailing	1	1	1	1	100%
Topic Planning – Generating Possible Topics	2	1	2	1	100%
Client Interviews and Topic Proposals	3	2	3	2	100%
Preparing and Presenting Topic Proposals (PPT)	5	1	5	1	100%
Getting Signatures for Proposals and Creating LOI	6	1	6	2	100%
Paper Adjustments and LOI Finalization	7	1	7	1	100%
New Client Search, New Topics, and UST Client Letters	8	1	8	2	100%
Interviewing New Clients and Looking for Technical Adviser	9	2	9	2	100%
THS1-Form2: Advisorship Agreement	10	1	10	1	100%
Chapter 1 – Company Background and Context	10	3	10	3	100%
Chapter 2 – Related Literature and Studies	10	3	10	3	100%
Chapter 2 – Related Systems and Technologies	11	3	11	3	100%
Chapter 2 – Synthesis	13	1	13	1	100%
Chapter 3 – Methodology & ETL	11	2	11	2	100%
Chapter 3 – Presentation of Star Schema	12	2	12	2	100%
Chapter 3 – Business Analytics Testing and Modeling	12	2	12	2	100%
Chapter 3 – Risk Assessment	13	1	13	1	100%
Mid-draft TA/CA Review (Pre-oral Check)	12	1	12	1	100%
Mock Defense	15	1	15	1	100%
Post Mock Revision	15	1	15	1	0%
Full Document Compilation and Review	13	1	13	1	0%
Safe Assignment Upload (UST Cloud Campus)	13	1	13	1	0%
Endorsement Form Filing (THS1-Form3A)	14	1	14	1	0%
Panel Availability Confirmation (THS1-Form4A)	14	1	14	1	0%
3 Manuscript Copies + Envelope Prep (Sec. 13.2)	14	1	14	1	0%
Defense Reminder to Panel (THS1-Form5A)	14	1	14	1	0%
Final Oral Defense – IS 26312	15	1	15	1	0%
Post-Defense Revisions (10-day window)	15	2	15	2	0%
Revised Document Submission (PDF + Ring-bound)	17	1	17	1	0%

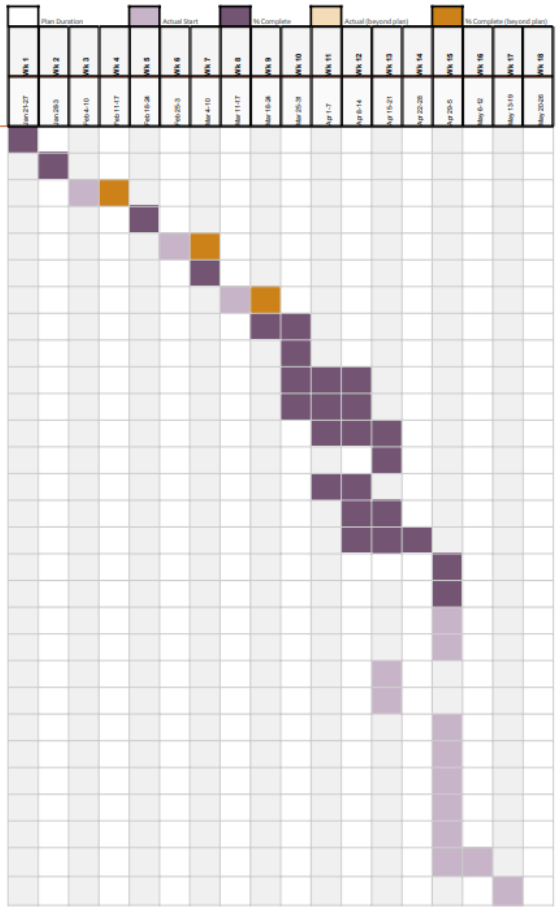


Figure 2. *Gantt Chart: Project milestones and timeline for IS 26312.*

IS 26316 – Capstone Project 2

Group 8 | Section 4ISA

ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
Chapter 1-3 Revisions from CP1 Defense	1	2	1	2	0%
Data Gathering and ETL Process	2	4	2	4	0%
Star Schema Design and Presentation	3	3	3	3	0%
BA Model Development and Testing	5	4	5	4	0%
BA Tools, Techniques, and Applications	6	3	6	3	0%
Risk Assessment and Mitigation	7	2	7	2	0%
Chapter 4.1 – Presentation of Results	8	3	8	3	0%
Chapter 4.2 – Data Analysis	9	3	9	3	0%
Chapter 4.3 – Dashboard Development	8	4	8	4	0%
Chapter 4.4 – Summary of Findings	11	2	11	2	0%
Chapter 5 – Conclusions and Recommendations	12	2	12	2	0%
Full Document Compilation (Ch 1-5)	13	2	13	2	0%
Mid-draft TA/CA Review	13	1	13	1	0%
Mock Defense	14	1	14	1	0%
Post-Mock Revisions	14	1	14	1	0%
Safe Assignment Upload (UST Cloud Campus)	14	1	14	1	0%
THS2-Form3B: Endorsement for CP Defense	15	1	15	1	0%
THS2-Form4B: Panel Availability Confirmation	15	1	15	1	0%
THS2-Form5B: Defense Reminder to Panel	15	1	15	1	0%
3 Manuscript Copies + Envelope Prep (Sec. 13.2)	15	1	15	1	0%
Final Oral Defense – IS 26316	16	1	16	1	0%
Post-Defense Revisions (10-day window)	16	2	16	2	0%
Final Document Submission (PDF + Book-bound)	18	1	18	1	0%

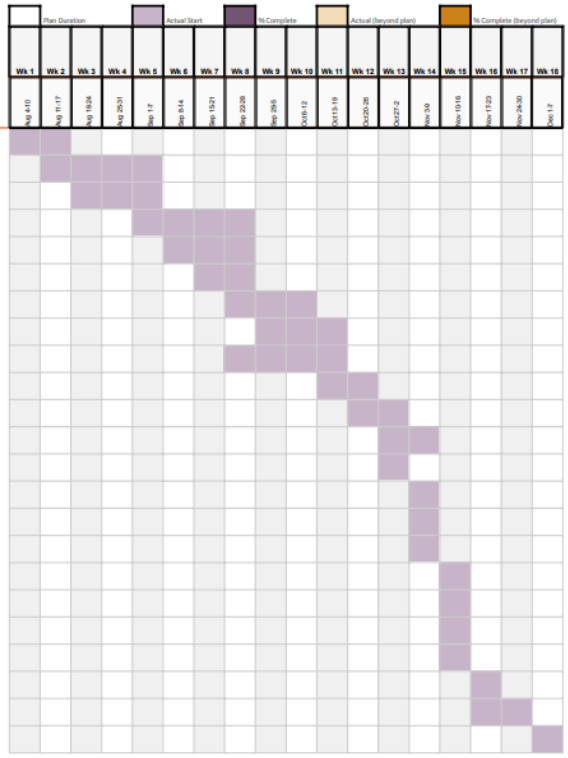


Figure 3. *Gantt Chart: Project milestones and timeline for IS 26316.*

Chapter 2: Review of Related Literature, Studies, Systems, and Technologies

2.1 Review of Related Literature and Studies

This section presents a thematic review of the literature and studies that establish the theoretical and methodological foundation for the proposed Demand Forecasting and Prescriptive Inventory Management Dashboard. The review is organized into five distinct areas. First, it explores the transition toward data driven inventory management and the asymmetric costs of forecasting errors in retail operations. Second, it examines the operational consequences of manual record keeping and the necessity of digitized data baselines. Third, it discusses Fast Slow Non moving classification as a mechanism for prioritizing high velocity merchandise. Fourth, it evaluates time series forecasting methods, including Prophet, ARIMA, exponential smoothing and intermittent-demand methods, as candidates for the USTore's seasonal and intermittent demand. Finally, it reviews the application of Business Intelligence dashboards as prescriptive decision support tools. Together, this body of literature validates the selected methodologies and provides the academic context required to address the operational challenges currently experienced by the USTore.

2.1.1 Inventory Management and Demand Forecasting in Retail

The global shift toward data-driven inventory management has fundamentally altered how retail organizations anticipate and respond to consumer demand. Traditional inventory replenishment methods, which relied on periodic physical counts and the experience-based judgment of store managers, have been widely documented as insufficient for managing the complexity and velocity of modern retail supply chains (Ugochi et al., 2024). Demand forecasting, defined as the practice of using historical sales data and statistical models to project future product requirements has emerged as the central operational capability that separates high-performing retail operations from those prone to chronic stockouts and excess inventory (Chowdhury et al., 2024; Liu et al., 2025).

In Philippine retail contexts, the challenge of implementing demand forecasting is compounded by the dominance of small-to-medium enterprises that lack the technical infrastructure and data literacy required to operationalize standard forecasting methodologies (Osio et al., 2026). Studies examining retail digitization in developing Southeast Asian markets consistently identify manual record-keeping as the primary bottleneck preventing the development of clean, complete data baselines; and without a reliable historical record, even sophisticated forecasting algorithms produce unreliable outputs (Boyd, 2025). The USTore's operational environment exemplifies this challenge, where a regulatory constraint against automated POS systems has prevented the accumulation of the itemized, date-stamped transaction history that demand forecasting requires.

Literature on retail inventory optimization further establishes that the costs of poor demand forecasting are asymmetric, though the relative magnitude varies depending on product and strategic context. (Park et al., 2025). Stockouts impose direct revenue losses and customer dissatisfaction effects that are difficult to recover, while overstocking generates carrying costs, obsolescence risk, and capital tied up in non-productive assets (Szabłowski, 2026). For consignment-based retail operations in particular where the store does not own the inventory outright; the consequences of inaccurate stock visibility are amplified because the timing and

accuracy of supplier payments depend directly on the reliability of sales records (Malhotra et al., 2017; Gerchak, 2002)

The operational vulnerabilities described in this literature directly mirror the current state of the USTore. Because the store operates without a reliable transaction history, it suffers from the exact asymmetric costs identified by Park et al. (2025), specifically recurrent stockouts of highly demanded apparel and immobilized capital tied up in stagnant items. Furthermore, the reliance on accurate batch sales reports for merchandise compensation aligns with the documented challenges of consignment retail.

The capstone project proposal that will encompass Demand Forecasting and Prescriptive Inventory Management Dashboard resolves these gaps by replacing experience based guesswork with a structured digital baseline, enabling the store to

operationalize the exact data driven forecasting practices that the literature identifies as essential for high performing retail environments.

2.1.2 Manual Record-Keeping and Its Operational Consequences

Despite the widespread availability of digital point-of-sale and inventory management solutions, a significant proportion of Philippine retail establishments continue to rely on paper-based sales recording systems (Anthique et al., 2025; Jacob, 2024). Non-adoption of digital systems is often linked to structural barriers such as regulatory uncertainty, infrastructure limitations, and financial constraints rather than a lack of awareness (Nirmani, 2025). In university-based retail environments, additional restrictions imposed by institutional financial governance frameworks further constrain the deployment of external transaction processing systems, even where the technology is commercially available (Islam, 2025).

Research on human error in manual data entry consistently demonstrates that error rates increase proportionally with transaction volume and time pressure (Bradley et al., 2026) In environments characterized by episodic high-traffic events such as enrollment periods and campus events in a university store; the conditions for manual recording failure are structurally embedded in the operation. Errors in manual tallying do not merely produce inaccurate historical records; they cascade forward into every downstream process that depends on those records, including stock replenishment decisions, supplier payment calculations, and cash flow projections (Reason, 2002).

Rather than full automation, literature points toward decoupled digital solutions that separate operational data capture from regulated financial processing. These lightweight systems enable structured recording of sales events without violating fiscal constraints, representing a pragmatic middle ground between manual processes and fully integrated POS infrastructures (Röding et al., 2023; Bouchard et al., 2023). Such tools are classified as internal inventory tracking systems rather than point-of-sale systems, preserving regulatory compliance while establishing the clean data baseline necessary for analytical processing. The proposed Digital Tallying Interface for the USTore is designed on precisely this principle, and its design rationale is grounded in the operational constraints and remediation strategies documented in this body of literature.

2.1.3 FSN Analysis and Inventory Classification

Fast-Slow-Non-moving (FSN) Analysis is a quantitative inventory classification method that segments a product catalog by sales velocity; the rate at which individual items are depleted from available stock over a defined period (Syntetos et al., 2014, Axsäter, 2006). The method assigns each Stock Keeping Unit (SKU) to one of three categories: fast-moving items that experience high and consistent turnover, slow-moving items that sell at irregular or reduced rates, and non-moving items that have recorded no sales activity within the observation window (Han et al., 2022). FSN Analysis is operationally significant because it enables inventory managers to allocate attention and replenishment resources proportionally to the items that drive the majority of store revenue (Tambunan et al., 2018).

FSN Analysis is closely related to the broader ABC Inventory Classification methodology, which similarly segments inventory by importance but ranks items by total value contribution rather than sales velocity (Silaen, 2024). The Pareto Principle, which is commonly expressed as the observation that approximately 80 percent of a system's outputs are produced by 20 percent of its inputs, provides the theoretical foundation for both classification approaches (Dunford et al., 2014). In a retail context, the Pareto Principle implies that a relatively small subset of the product catalog will account for the majority of sales volume, and that disproportionate stockout risk is concentrated in that subset (Hickins, 2023) For the USTore, where popular apparel items and institutional memorabilia follow distinctly seasonal demand patterns driven by the academic calendar, FSN classification is the necessary first step in identifying which items require proactive demand forecasting and which can be managed with simpler replenishment heuristics.

Studies applying FSN and ABC classification in university bookstore and institutional retail environments report that classification alone, even prior to formal forecasting model development produces measurable improvements in restocking accuracy by directing management attention toward the high-velocity segment of the catalog (Mohammed et al., 2024). The classification output also serves a dual function in the proposed system architecture as it determines which items are included in the time-series forecasting model and which items are managed through simple threshold rules, reducing computational overhead and improving overall system performance.

2.1.4 Time-Series Forecasting Methods in Retail Analytics

Time-series forecasting methods model historical data sequences as a function of time and project future values based on the patterns, trend, seasonality, and residual noise identified in the historical record (Kolambe, 2024). In retail inventory management, time-series forecasting is applied to historical sales or depletion data to generate demand estimates for defined future periods, which then serve as the basis for Reorder Point and Safety Stock calculations (Seyedan et al., 2023). The accuracy of these demand estimates is typically evaluated using a complementary suite of error metrics. Mean Absolute Percentage Error (MAPE) expresses forecast deviation as a percentage of actual demand, providing a scale-independent measure that enables comparison across SKUs with different sales volumes. Mean Absolute Error (MAE) reports the average forecast deviation in the same physical units as the original data, offering an operationally intuitive measure of how many units a forecast typically misses by. Root Mean Square Error (RMSE) applies a squared penalty that weights large forecast errors disproportionately, capturing the risk of extreme deviations that could trigger stockouts (Hyndman & Athanasopoulos, 2018). Together, these three metrics directly determine the reliability of the restocking advisories the system produces.

Two forecasting approaches are most widely applied in retail analytics research: the Autoregressive Integrated Moving Average (ARIMA) family of models and the Prophet algorithm developed by the Meta Platforms research team (Devi et al.,

2025, Mustapha et al., 2025). A third body of work addresses intermittent demand, where many periods record no sales: Croston's method (Croston, 1972), its bias-corrected variant SBA (Syntetos & Boylan, 2005), and the TSB method (Teunter, Syntetos & Babai, 2011), which lets the forecast decay when an item stops selling. Simple averages are often hard to beat and serve as baselines (Hyndman & Athanasopoulos, 2018). ARIMA models, which decompose a time series into autoregressive, differencing, and moving average components, perform well on stationary or near-stationary series with clear linear trend and limited seasonality complexity (Szostek et al., 2024). However, ARIMA models require a substantial historical record, typically a minimum of two to three years of consistent, clean observations (50-100) data points to produce reliable parameter estimates, and their performance degrades significantly when the series contains multiple overlapping seasonal cycles (Hyndman & Athanasopoulos, 2018). Exponential smoothing methods, particularly the Holt-Winters family, represent another established approach to time series forecasting that incorporates level, trend, and seasonal components through exponentially weighted moving averages (Hyndman et al., 2002). Holt-Winters has been widely adopted in retail and supply chain contexts due to its computational simplicity and strong performance on series with a single well-defined seasonal cycle. However, the standard Holt-Winters formulation is limited to modeling one seasonal period per series, which makes it unsuitable for the USTore's demand environment where sales are simultaneously influenced by weekly patterns, semester-level cycles, and irregular academic calendar events (Hyndman & Athanasopoulos, 2018). While extensions of exponential smoothing to handle multiple seasonality exist, these require

substantially more complex state space formulations that negate the method's core advantage of simplicity without offering the same flexibility for user-specified holiday effects and structural changepoints that Prophet provides natively.

Prophet, by contrast, is designed specifically for business time series that exhibit strong multiple seasonality patterns, irregular holiday effects, and structural changes which are characteristics that are directly applicable to the USTore's demand environment, where sales are systematically influenced by the academic semester calendar, enrollment events, university foundation days, and semestral breaks (Saeed et al., 2022). Prophet fits an additive decomposition model with separate components for trend, multiple seasonality levels, and user-specified holiday effects, and it is demonstrably robust to missing data and outliers introduced by periods of low or zero transaction volume (Brownlee, 2020). For datasets with limited clean history, such as the USTore's post-digitization records, Prophet's Bayesian framework allows for informative prior specification that compensates partially for data scarcity, making it a strong candidate for this deployment context.

Industry benchmarking studies evaluating Prophet against ARIMA in retail and e-commerce demand forecasting contexts consistently report that Prophet achieves competitive or superior MAPE scores on seasonal retail series while requiring substantially less manual tuning of model parameters (Srinivasan et al., 2025). A MAPE threshold of 20 percent or lower is widely cited in retail forecasting literature as the practical accuracy benchmark that justifies algorithmic-driven replenishment decisions over manual estimation; below that threshold, the forecast is sufficiently reliable to inform Reorder Point calculations without significant manual override

(Kumar et al., 2024). However, MAPE is undefined when actual sales are zero and unstable when they are small, which is common for individual retail items (Hyndman & Koehler, 2006).

The comparative literature identifies Prophet, ARIMA, exponential smoothing and intermittent-demand methods as candidates for highly seasonal, event-driven business data; the model used in the proposed system is selected by comparing these candidates on the USTore's own sales data (Section 3.3.4). Because the USTore experiences massive demand spikes tied to specific academic calendar events, ARIMA models would fail to capture these irregular seasonalities. Furthermore, the project targets the established 20 percent MAPE threshold for the selected forecasts and reports the result whether or not it is met, while the Reorder Points and Safety Stock levels are calculated from each item's actual sales over the last 365 days and judged separately against the USTore acceptance standard (Section 3.3.4).

2.1.5 Dashboard-Based Decision Support in Retail and Institutional Operations

Business Intelligence (BI) dashboards have emerged as the preferred output layer for analytics systems deployed in retail management contexts because they translate statistical outputs which require data science literacy to interpret into visual formats accessible to operational decision-makers without technical backgrounds (Ajax et al., 2025). In inventory management applications specifically, dashboards that incorporate color-coded alert systems, automated report generation, and real-time or near-real-time stock status visualization have been shown to reduce the cognitive load

on store managers and accelerate replenishment decision cycles (Rahman et al., 2025).

Microsoft Power BI and Tableau are the two most widely adopted enterprise BI visualization platforms in both commercial and academic research contexts. Power BI is distinguished by its deep integration with the Microsoft Office and Azure ecosystem, its accessible free desktop license tier, and its well-documented capability to generate publication-quality PDF exports, which is a functional requirement for organizations that use standardized report formats for internal financial documentation (Zamil et al., 2024). A study by Yerra (2025) found that Power BI dashboard implementations in inventory management contexts produced average improvements of 85 percent in inventory accuracy and 92 percent in demand forecast reliability, attributing these gains to the shift from periodic manual reporting to continuous automated monitoring.

In Philippine university administrative environments, the prevalence of Microsoft Office tooling across finance, purchasing, and administrative departments creates a meaningful practical advantage for Power BI over Tableau, as Power BI reports are natively compatible with the document workflows already in use by the UST Purchasing Office and Finance Department. The dashboard proposed for the USTore is designed to serve both operational users, store management staff who monitor stock status, and administrative users, the Purchasing Office and Finance Department who require accurate batch sales reports for supplier compensation within a single unified visualization environment.

2.1.6 Synthesis

The literature reviewed in this selection collectively reinforces the academic foundation for the proposed capstone research. Each subsection addresses a distinct layer of the problem and the solution for it, and they trace a continuous logical thread from the operational problem through the methodological response.

The most important literature review is the relationship between data quality and forecasting reliability. Ugochi et al. (2024), Chowdhury et al. (2024), and Boyd (2025) collectively establish that demand forecasting is only as reliable as the transaction history on which it is trained, while Bradley et al. (2026) and Reason (2002) independently confirm that manual recording systems structurally produce the kind of incomplete, error-prone data that makes reliable forecasting impossible. These research studies, drawn from separate disciplines of supply chain management and human error research respectively, conclude on the same idea: that the USTore's forecasting problem cannot be solved at the modeling layer without first being solved within the data capture layer. This cross-disciplinary convergence is what justifies the Digital Tallying Interface not as a peripheral feature of the proposed system but as its most foundational component, because without it, every analytical method reviewed in sections 2.1.3 and 2.1.4 would operate on a corrupted input.

Next is the significance between the FSN Analysis literature and the time-series forecasting literature. Tambunan et al. (2018), Mohammed et al. (2024), and Syntetos et al. (2014) establish that FSN classification improves restocking accuracy by directing analytical resources toward the high-velocity segment of a

catalog, while Hyndman and Athanasopoulos (2018), Saeed et al. (2022), and Srinivasan et al. (2025) establish that Prophet outperforms ARIMA specifically on seasonal, event-driven series. When these findings are combined, a design logic emerges that neither body of literature concludes on its own: FSN classification is not merely a descriptive tool but a necessary pre-processing step that makes item-level demand forecasting computationally tractable and methodologically appropriate. By forecasting item-level demand only for fast-moving items identified through FSN analysis, the system concentrates forecasting resources on the exact subset of the catalog where seasonal demand patterns are strongest and where the consequences of forecasting error, stockouts of high-demand merchandise, are most severe.

Lastly, the most practically significant for the USTore's specific institutional context, which emerges between the consignment retail literature and the dashboard usability literature. Malhotra et al. (2017) and Gerchak (2002) establish that consignment arrangements amplify the consequences of inaccurate sales records because supplier compensation is directly tied to recorded sales volume, while Vázquez-Ingelmo et al. (2019), Rahman et al. (2025), and Yerra (2025) establish that BI dashboards with automated report generation and non-technical user interfaces are the mechanism through which analytical outputs are converted into operational and administrative action. By combining these two findings it reveals that the Power BI dashboard's automated monthly batch sales report function is not simply a convenience feature but the specific point in the system architecture where the forecasting and classification work is translated into the supplier compensation

accuracy that the consignment model demands. The dashboard is not the output of the analytics; it is the bridge between the analytics and the store's financial obligations.

When the reviewed literature and systems are read in conjunction rather than in isolation, it becomes clear that the proposed system's value does not reside in any single component but in the combination and integration of all four layers. Each component of the proposed architecture is individually supported by its respective body of literature: the Digital Tallying Interface by lightweight data capture research in compliance-constrained environments, the FSN module by velocity-based inventory classification studies, the forecasting module by comparative forecasting studies on seasonal business time series and intermittent-demand research, and the Power BI dashboard by BI implementation research on non-technical user accessibility. The integration logic connecting these components, which is detailed in Section 2.1.6, is what elevates the system from a collection of individually justified tools into a coherent analytical pipeline.

2.2 Review of Related Systems and Technologies

The following section explores existing systems and technologies deemed relevant to the proposed system for the USTore. Included in the review is the evaluation of inventory management and Enterprise Resource Planning Systems (ERP). The evaluation establishes the constraints that render the mentioned systems as unviable for the operations of the USTore due to compliance with BIR regulations. Other systems that aim to address these constraints are digital tallying and data

capture systems which contain relevant features in the development of the proposed system. Technologies such as Python tools are evaluated due to the important frameworks that assist in data processing and the intent to conduct demand forecasting and prescriptive analytics. The use of Power BI as the visualization tool for processed data and analyses is reviewed due to its accessibility and cost effectiveness.

2.2.1 Existing Inventory Management and POS Systems

Commercial inventory management systems such as Lightspeed Retail, Vend, and TradeGecko represent the current state of the art in cloud-based, small-to-medium retail inventory platforms. These systems integrate POS transaction processing, real-time stock tracking, supplier management, and demand forecasting into unified platforms delivered via web interfaces (Lightspeed, n.d). Their functional scope significantly exceeds the requirements of the USTore deployment context; however, they are categorically inapplicable because they operate as point-of-sale systems that process monetary transactions; placing them in direct conflict with the BIR compliance constraint and university policy governing the USTore's operations.

Enterprise Resource Planning (ERP) systems with integrated inventory modules including Oracle NetSuite, SAP Business One, and the Microsoft Dynamics 365 suite offer more comprehensive supply chain management capabilities that include automated reorder generation, inventory replenishment management, and advanced forecasting engines. Institutional ERP deployments in Philippine universities

have been documented in the academic literature, with the University of Santo Tomas itself operating administrative systems under enterprise licensing agreements (Jenkins, 2026). However, ERP-level solutions impose substantial implementation complexity, licensing cost, and institutional IT governance overhead that are disproportionate to the USTore's operational scale and technical support capacity. The gap between what enterprise systems offer and what the USTore's constrained environment can accommodate defines the design space the proposed system occupies.

2.2.2 Non-Transactional Digital Tallying and Lightweight Data Capture Systems

In compliance-constrained retail environments where full POS automation is impermissible, researchers and practitioners have documented the deployment of lightweight, non-transactional digital data capture tools as an intermediate solution for improving inventory data quality. Google Forms connected to Google Sheets back ends has been widely adopted in small Philippine enterprises as a zero-cost, institutionally acceptable data collection mechanism that requires no server infrastructure and produces structured, queryable output (Davis et al., 2023). Studies evaluating the accuracy improvement achieved by transitioning from paper-based tallying to structured digital form submission consistently report significant reductions in missing-entry and transcription error rates, even in the absence of full POS integration (Tate et al., 2021).

Web application frameworks including Flask and React have been applied in academic capstone and research projects to develop customized, institution-specific data capture interfaces that go beyond the configurability of generic form tools. Flask-based lightweight web applications deployed in Philippine institutional contexts demonstrate that functional inventory logging tools can be developed and maintained without enterprise-grade infrastructure, using free open-source components and minimal server requirements. The proposed Digital Tallying Interface for the USTore draws directly from this body of work in its design philosophy: a purpose-built, lightweight, compliance-preserving data entry tool that prioritizes data structure and completeness over feature richness.

2.2.3 Python-Based Analytics Frameworks for Inventory and Demand Forecasting

Python has become the dominant programming language for data analytics, machine learning, and statistical modeling in both research and applied contexts, owing to its extensive ecosystem of purpose-built libraries, its accessibility to practitioners without formal computer science training, and the breadth of community-generated documentation supporting its application in business analytics workflows (Singh, 2025). In inventory management analytics specifically, the pandas library provides the foundational data manipulation and transformation capabilities required for ETL processing, including the computational operations necessary for the FSN classification module such as the calculation of Average Daily Units Sold per SKU, the derivation of Pareto-based percentile thresholds across the product catalog,

and the assignment of velocity categories through rule-based conditional logic applied directly to the structured sales dataset

Published capstone and research projects applying Python-based analytics pipelines to Philippine retail and supply chain contexts confirm that the language's toolchain is appropriate and functional in resource-constrained deployment environments (Oguis et al., 2022). The Star Schema dimensional data model, implemented via structured CSV or SQLite in lightweight deployments, has been documented as a viable intermediate data warehouse architecture for small-scale BA projects where full enterprise data warehousing infrastructure is unavailable (Kimball & Ross, 2002). The combination of Python for back-end analytics and a dimensional model for data storage represents a well-established open-source analytics stack with a proven track record in academic and small-business BA deployments.

2.2.4 Power BI Dashboard Implementations in Inventory and Retail Contexts

Power BI dashboard deployments in inventory management contexts have been the subject of growing applied research interest, particularly as organizations seek to make analytics outputs accessible to non-technical operational users without the mediation of data specialists. Implementations documented in the literature cover a range of retail and supply chain applications including stock level monitoring, supplier performance tracking, sales trend visualization, and automated report generation for financial reconciliation purposes (Garudasu et al., 2025). The common finding across these implementations is that the translation of analytical outputs into interactive

dashboard visualizations accelerates replenishment decision cycles and reduces dependency on manual periodic reporting.

The PDF export functionality native to Power BI has been specifically documented as a critical feature for organizations operating in environments where official financial reports must be submitted in standardized document formats, a requirement directly applicable to the USTore's obligation to produce monthly batch sales reports for the UST Purchasing Office and Finance Department (Yerra, 2025). Case studies of Power BI implementation in Philippine institutional and government contexts further confirm that the platform's free desktop license and Microsoft ecosystem compatibility make it accessible to organizations without dedicated BI infrastructure budgets (Shaik, 2024).

2.2.5 Synthesis

The systems and technologies reviewed in this section span from functional sophistication, from enterprise-grade ERP platforms to open-source analytics libraries to lightweight form-based data capture tools. These four related systems and technologies reveal that sophistication alone does not determine suitability. For businesses operating under BIR compliance constraints, strict university financial governance, and limited technical infrastructure, the determining criterion for all of this is the strategic combination of technologies that satisfy these regulations while delivering the high-level predictive modeling that off-the-shelf platforms cannot. The most important insight emerges from Sections 2.2.1 and 2.2.2 against each other.

Commercial POS and ERP systems possess the forecasting capabilities that the USTore requires, but their fundamental architecture as a transactional system make them categorically inapplicable under BIR regulations. Conversely, lightweight non-transactional tools satisfy the compliance constraint but offer no analytical depth beyond simple data collection. This tension defines the “design gap” that the proposed system occupies and solves: a

compliance-preserving data capture layer that feeds a separate, high-powered analytics engine. This architecture achieves both regulatory safety and predictive power without sacrificing one for the other.

The second insight surfaces by reading the Python analytics frameworks in Section 2.2.3 alongside the Power BI documentation in Section 2.2.4. The Star Schema structure produced by the Python ETL pipeline is not merely a storage convention but the specific data format that Power BI is optimized to query. This means the FSN categorization, demand forecast, and Reorder Point outputs are stored at item level as direct inputs for the dashboard's threshold alerts and advisories rather than as standalone analytical results.

Lastly, Power BI's PDF export capability, documented by Yerra (2025) and Garudasu et al. (2025), takes on heightened significance in the USTore's consignment context. Because the UST Purchasing Office and Finance Department require standardized monthly batch sales documents to execute supplier compensation, PDF

export is not a convenience but a functional necessity that determines whether the system's outputs can actually be operationalized.

2.3 Chapter Synthesis

The body of literature and systems reviewed in this chapter collectively establishes three converging lines of support for the proposed Demand Forecasting and Prescriptive Inventory Management Dashboard. The first line of support addresses the foundational problem. Research on inventory management in retail and Philippine SME contexts confirms that manual, assumption-based inventory practices generate compounding operational inefficiencies, distorted stock baselines, unreliable demand signals, and reactive replenishment cycles that directly translate into stockout events and revenue losses. This finding is documented across both global supply chain literature and Philippine retail digitization studies, validating the operational problem identified in the USTore's current state and establishing the academic basis for the proposed system's intervention.

The second line of support addresses the solution architecture. The reviewed literature on non-transactional data capture systems, Python-based analytics frameworks, FSN inventory classification, and time-series and intermittent-demand forecasting provides direct academic and applied precedent for each component of the proposed system's design. The Digital Tallying Interface is grounded in documented lightweight data capture approaches used in compliance-constrained environments. The FSN classification module is supported by inventory management literature establishing velocity-based segmentation as the appropriate first-order classification method for retail catalogs with diverse turnover profiles. The forecasting module is supported by comparative forecasting studies on seasonal business time series and by intermittent-demand research on items that record no sales on most days (Teunter, Syntetos & Babai, 2011). The Power BI dashboard layer is supported by BI implementation research confirming its accessibility to non-technical users and its functional suitability for automated report generation.

The third line of support addresses the system gap. A review of existing commercial inventory management and POS systems confirms that no currently available off-the-shelf solution can be deployed in the USTore's constrained institutional environment without violating BIR compliance requirements or exceeding the organization's technical infrastructure capacity. The proposed system occupies a design space; lightweight, compliance-preserving, analytics-capable, and dashboard-accessible that existing solutions do not address. This gap validates the contribution of the proposed project and confirms that its development is not

duplicative of existing work but represents a contextually specific application of established analytical methods to an underserved institutional retail context.

Taken together, the literature and system reviews confirm that the proposed project is methodologically grounded, technically feasible within the stated constraints, and responsive to a documented and unaddressed operational need. The design decisions formalized in Chapter 3 are each traceable to findings and precedents established in this chapter's review.

Chapter 3 : Methodology

The development of the Demand Forecasting and Prescriptive Inventory Management Dashboard follows the Cross Industry Standard Process for Data Mining (CRISP DM). This framework provides a structured, iterative approach to business analytics projects, ensuring that the analytical models remain strictly aligned with the operational requirements of the USTore. The methodology is structured across six phases, executed iteratively to allow model refinement when performance criteria are not met.

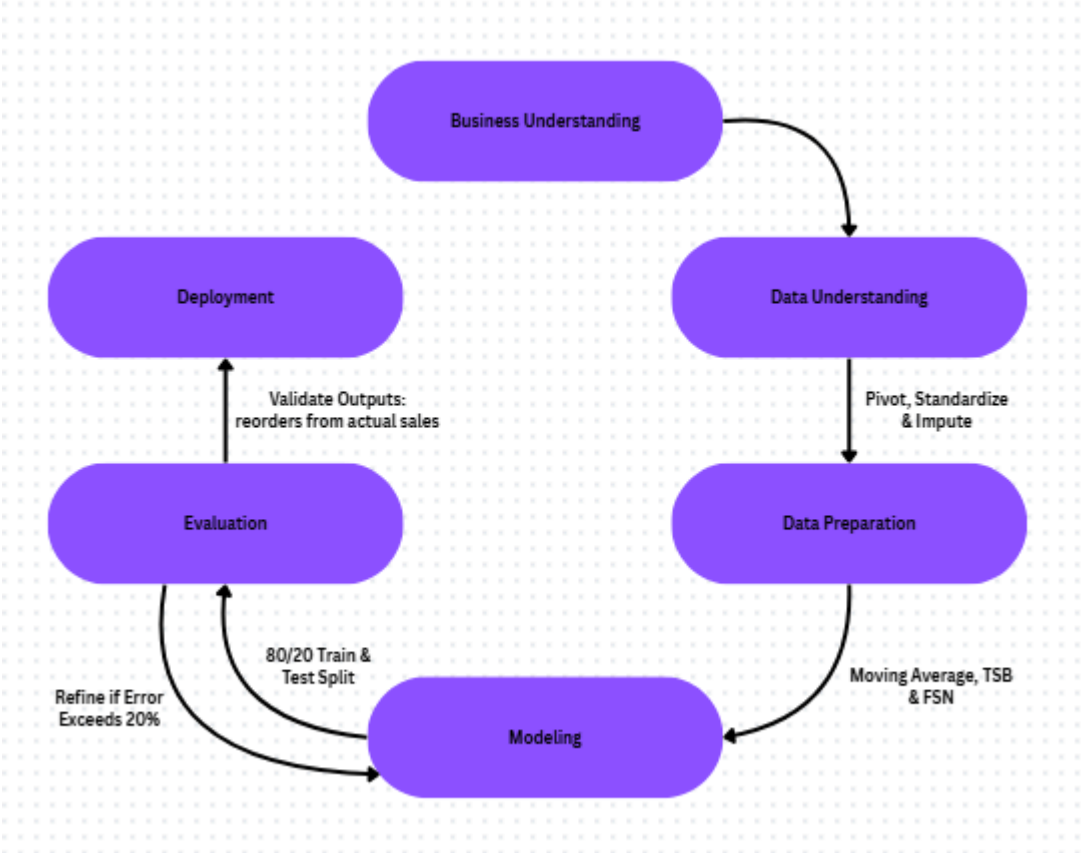


Figure 3. *CRISP-DM Framework Applied to the USTore Demand Forecasting Project*

Business Understanding

This initial phase defines the operational challenges and project objectives. The USTore's primary challenge is its reliance on manual tallying methods, which distorts the inventory baseline and causes recurrent stockouts that can result in lost sales revenue. The objective is to transition to a data backed model by prescribing optimal inventory replenishment policies. By incorporating Fast Slow Non moving classification, the store can prioritize its highest velocity merchandise, focusing

forecasting efforts on profitable items to actively prevent stockouts and maximize sales.

Data Understanding

This phase focuses on identifying and assessing the available data sources. The project relies on historical Excel sales tallies starting from the 2023 baseline, supplementary daily sales reports, initial inventory count sheets, and the official university academic calendar. During this assessment, critical data quality challenges were identified, including irregular episodic recording dates, inconsistent item descriptions, and price grouped entries lacking item level specificity.

Data Preparation (ETL Process)

The third phase addresses the data quality issues identified above. As detailed in the Extraction, Transformation, and Loading sections, the raw Excel records undergo a structural pivot, standardized naming correction, and proportional imputation. The clean, validated dataset is then deposited into a Star Schema dimensional data model within a SQLite database, readying it for analytical processing.

Modeling (Business Analytics)

In this phase, statistical and mathematical algorithms are applied to the prepared data. The project employs a multi-layered modeling approach: Fast Slow Non moving Analysis to segment the catalog by sales velocity; a six-month moving average per category and a blend of category share and TSB per fast-moving item, both adjusted for the academic calendar, to generate 30-day demand forecasts; and

prescriptive mathematical formulas to calculate specific Reorder Points, Safety Stock thresholds, and Economic Order Quantities.

Evaluation

Before deployment, the forecasting models must be evaluated against strict performance criteria to ensure their prescriptive outputs are reliable. The forecasting models undergo walk-forward validation using an 80 and 20 train-test split on the historical data. The primary acceptance criterion is a Mean Absolute Percentage Error (MAPE) of 20 percent or lower, supplemented by Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) as diagnostic metrics reported per SKU and per category against a naive baseline. Models failing the MAPE threshold are retuned and compared with alternative methods; a threshold still not met is reported, not tuned away.

Deployment

The final phase translates the validated mathematical models into an accessible operational tool. The structured database and analytical outputs are integrated into a Microsoft Power BI dashboard. This deployment layer features color coded safety stock alerts and automated PDF export capabilities, delivering actionable, prescriptive insights through calendar-contextual interpretation cards directly to USTore management and the university finance channels.

3.1 Data Gathering (with ETL)

Data gathering for the project encompasses both the collection of all relevant source data and its subsequent preparation for analytical use through a structured Extract, Transform, and Load (ETL) pipeline. Given the USTore's reliance on manual tallying methods and the resulting inconsistencies in its historical records, raw data cannot be used directly for modeling without first undergoing systematic cleaning and restructuring. This section documents the four components of the data gathering process: the identification of data sources, the extraction of raw records from each source type, the transformation operations applied to resolve quality issues, and the loading of the prepared dataset into the project's dimensional data model. Together, these processes ensure that all inputs to the FSN classification, demand forecasting, and prescriptive inventory calculations are accurate, consistently structured, and traceable to their original source records.

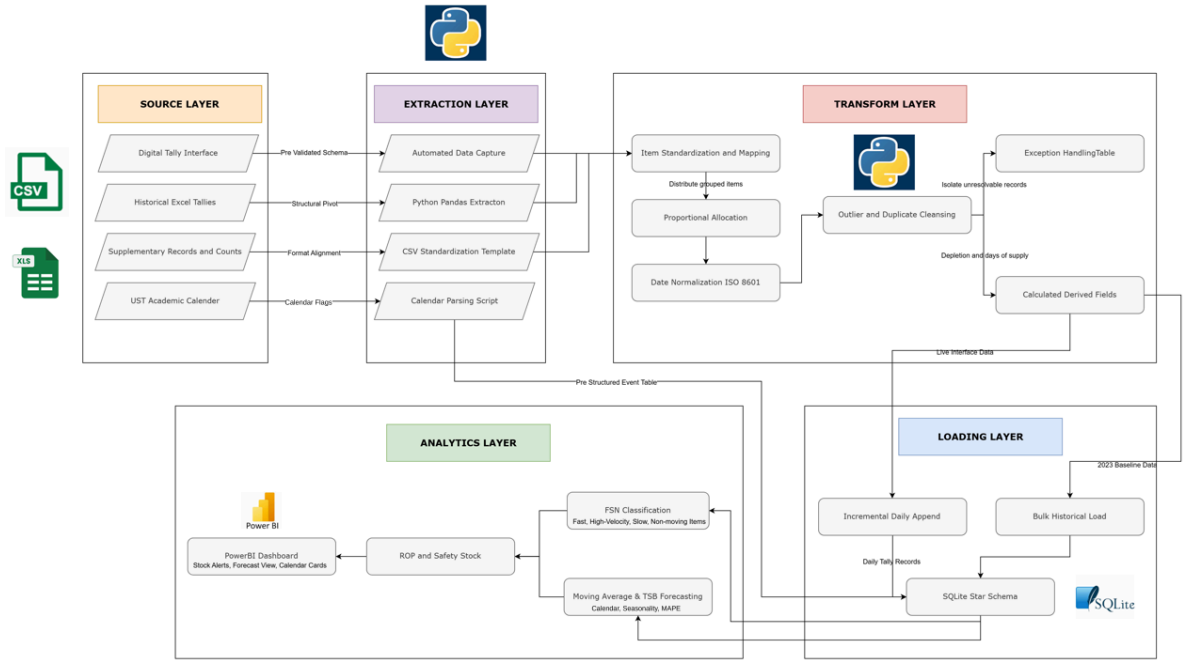


Figure 4. *Data Pipeline for the USTore Demand Forecasting and Prescriptive Inventory Management Dashboard*

3.1.1 Data Sources

The project will utilize three categories of data sources. The first category is the existing structured tally records currently maintained by the USTore in digital Excel format. These records consist of monthly batch sales tallies organized by consignment supplier, covering the period from 2023 through 2026. Each monthly file records transaction quantities per item per tally date, where tally dates represent discrete days on which sales entries were recorded rather than continuous daily observations. Collectively, the historical dataset will reflect the episodic recording pattern inherent in

the USTore's current manual tallying system, where each column in the spreadsheet corresponds to a specific recorded selling day within that billing month. Months vary in the number of tally dates captured, ranging from three recorded dates in some periods to as many as 29 in others, depending on transaction volume and operational circumstances during that month.

The second category consists of supplementary historical records maintained by the USTore in earlier formats. Specifically, a Daily Sales Report file covering May 2024 provides a more itemized format than the standard tally sheets, including unit prices, production costs, discounted price tiers, and per-item profit figures. While not directly compatible with the primary tally sheet structure, the record will be standardized and incorporated into the dataset as a supplementary baseline for price and cost validation during the ETL transformation phase. Additionally, the initial monthly inventory count sheets document on-hand stock quantities at the start of each billing month and will serve as the reference baseline for stock depletion rate calculations.

The third category involves new transactional data generated by the Digital Tallying Interface, which will be deployed at the start of the Capstone Project–2 implementation phase. Moving forward, the interface will produce structured, itemized daily sales records, capturing the item sold, quantity, transaction date, and transaction type for each sales period. The transaction type field is included, because it allows the system to distinguish confirmed sales from non-sale inventory removals, such as damaged goods or promotional items, ensuring that the demand forecast is based strictly on revenue-generating transactions. Furthermore, the interface will include a

dynamic holiday mapping feature. This feature will provide a UI toggle for “Store Closure/Suspension”, allowing the staff to be able to tag specific dates as non-operation periods. Dates that are tagged will be automatically updated in the Dim_Date table's is_store_closed field, which records planned store closures separately from market-driven demand drops. Once enough records are gathered, this live data will constitute the primary clean dataset used for time-series model training and validation. Meanwhile, the historical tally data will serve as the supplementary baseline for initial FSN classification and trend estimation.

A fourth, dynamic data category covers demand-influencing events not captured in the official UST academic calendar, such as college-specific celebrations, sports fests, acquaintance parties, and token seasons. The Digital Tallying Interface will include an event flagging feature that allows USTore staff to mark any date as an event day with a descriptive label. Flagged dates are recorded in a supporting operational table called Event_Log, which contains the event date, event name, event description, the staff member who logged it, and the date it was recorded. During each ETL cycle the pipeline reads Event_Log and updates the is_event_day flag in Dim_Date for the corresponding calendar dates, ensuring that the event flags remain current without requiring event-specific Boolean columns for each individual unscheduled event in Dim_Date. Separately, the interface's Store Closure/Suspension toggle updates the is_store_closed flag in Dim_Date, a dedicated Boolean that signals zero expected demand and is kept distinct from is_event_day so that planned closures are not confused with demand-influencing events.

An additional reference input required for the prescriptive inventory calculations is the estimated supplier lead time. This value represents the approximate number of calendar days between the placement of a restocking request and the arrival of replacement merchandise from the consignment supplier. The lead time estimate is not recorded in any of the USTore's existing Excel tally sheets or inventory count records. Instead, it is a generalized operational estimate relayed by USTore management based on their experience coordinating with consignment suppliers. Because the estimate is provided as a general figure rather than a per-item or per-supplier measurement derived from transactional data, it will be manually entered into the Dim_Product table as a uniform default value applied across all active SKUs. Should USTore management provide differentiated lead time estimates for specific suppliers in the future, the field can be updated on a per-product basis without requiring structural changes to the data model. As supplier-specific lead time observations become available during the Capstone 2 implementation phase, the lead_time_days field will be updated on a per-supplier basis to improve ROP calculation precision.

An additional set of reference inputs required for the Economic Order Quantity (EOQ) calculation consists of two cost parameters: the estimated ordering cost per replenishment cycle and the estimated holding cost per unit per period. The ordering cost represents the administrative expense incurred each time a restocking request is placed with a consignment supplier, including the labor cost of generating a Purchase Order through the UST Purchasing Office and coordinating delivery logistics. The holding cost represents the per-unit cost of storing unsold inventory over time,

including the opportunity cost of capital tied up in stock and any applicable storage or handling expenses. Because the USTore does not currently track these cost figures in its existing records, both values will be estimated in consultation with USTore management and the UST Purchasing Office during the Capstone 2 onboarding phase. The estimates will be entered as configurable parameters in the system so that they can be refined as more precise cost data becomes available.

An additional data source required for the forecasting component is the official University of Santo Tomas academic calendar for academic years covered by the data scope from 2023-2026. The calendar provides the dates of enrollment periods, midterm and final examination weeks, university foundation events, and semestral breaks, all of which constitute identifiable demand-influencing events; the enrollment-period, examination-week and semestral-break dates feed the academic-calendar adjustment of the demand forecasts (Section 3.3.2). Because the academic calendar is published and publicly accessible via the official University website, this dataset requires no additional access permissions beyond those already established with the USTore. For the post-deployment phase, the Digital Tallying Interface will serve as the primary source for identifying unplanned suspensions and store closures. This transitions from a static calendar to a dynamic one which will allow the forecasting model to achieve higher precision as the post-implementation dataset grows.

Table 2. Summary of Data Sources

Data Type	Source	Format	Period
Monthly beginning inventory counts	USTore	Excel/Sheet	2023-2026
Historical sales tally records	USTore	Excel/Sheet	2023-2026
Daily Sales Reports	USTore	Excel/Sheet	2023-2026
Clean daily sales (post-deployment)	Digital Tallying Interface	Structured digital entries (SQLite / Google Sheets)	From Capstone 2 deployment onward

Estimated supplier lead time	USTore Management (Verbal/informal)	Manually entered reference value	One-time input at project onboarding; updated as needed
Ordering and holding cost estimates	USTore Management and UST Purchasing Office (Verbal/informal)	Manually entered reference values	One-time input at project onboarding; updated as needed
Academic calendar events	UST Website and Digital Tallying Interface	Public document / PDF	2023-2026
Unscheduled event records	Digital Tallying Interface Event Flag Feature	SQLite Event_Log table	Ongoing; logged by USTore staff as events occur

Store closure and suspension flags	Digital Tallying Interface Holiday Mapping Toggle	is_store_closed field in Dim_Date	Ongoing; logged by USTore staff when closures or suspensions occur

3.1.2 Data Extraction

The extraction phase of the ETL pipeline will operate differently for each data source type. For the existing digital tally sheet records spanning 2023 to 2026, extraction will proceed through a structured parsing process: each monthly Excel file will be read programmatically using the Python pandas library, which will identify supplier section headers, item row labels, tally date column headers, and quantity values. Because the existing tally sheets use a wide-format layout with tally dates as column headers rather than rows, the extraction step will also perform a structural pivot, converting each file from wide format (one column per tally date) to long format (one row per item-date-quantity observation) before ingestion into the processing pipeline. A Data Gathering Log Sheet will be maintained throughout the extraction phase to document dates, responsible team members, and quantities of records processed.

For historical inventory count sheets and any tally records that feature incompatible layouts, extraction will proceed through a data standardization process. This proposed action supports the literature that structured entry reduces downstream

errors (Redman, 2001). These structurally inconsistent Excel files will be programmatically reformatted into a standardized CSV template that enforces a consistent column schema: date, item description, quantity, and record type before ingestion into the processing pipeline. For the May 2024 Daily Sales Report, extraction will follow the same standardization process, with additional fields captured for unit price and production cost where available to support cost validation.

For the live transaction data generated by the Digital Tallying Interface, extraction is automated: each completed tally entry is written directly to the project's SQLite database or Google Sheets back end in a pre-validated schema that eliminates the transcription error vector. The interface will enforce field-level validation, requiring non-null values for item, quantity, date fields and transaction type to prevent the entry of structurally incomplete records. For the academic calendar, extraction will involve parsing the official UST calendar document to produce a structured event table listing event names, start dates, and end dates, formatted for loading into the Boolean calendar flags of the Dim_Date table.

To mitigate encoding errors, the system will implement an intra-day completeness audit. Rather than waiting until the end of the business day to detect missing entries, the Digital Tallying Interface will trigger a "Data Pending" alert at a configurable mid-day checkpoint if no sales entry has been recorded for the current date. This alert will be visible both in the tallying interface itself and in the Power BI dashboard's data quality monitoring view. The purpose is to ensure that the system can distinguish between genuine zero-sales days, where the store was open but no transactions occurred, and encoding lapses, where sales occurred but were not

recorded. Days that pass the end-of-day cutoff with no entry and no staff confirmation of zero sales will be flagged as "Unverified Zero" in the Fact_Sales table rather than recorded as a confirmed zero. This flag will allow the forecasting model to treat unverified zeros as missing data rather than as true observations, preventing encoding lapses from suppressing the demand forecast. For historical records predating the Digital Tallying Interface, all zero-value entries will be treated as missing data by default, since there is no way to retroactively distinguish genuine zeros from recording omissions in the manual tally system.

For unscheduled event records logged through the Digital Tallying Interface event flagging feature, extraction is automated. Each event entry written to the Event_Log table is read by the ETL pipeline during each processing cycle. The pipeline extracts the event_date and event_name fields and maps them to the corresponding calendar_date rows in Dim_Date, updating the is_event_day flag to true for those dates. This ensures that the Dim_Date event flags are updated with newly identified demand-influencing events without requiring manual intervention in the data model. Store closures and suspensions flagged through the separate holiday mapping toggle follow a different path: they update the is_store_closed flag in Dim_Date rather than is_event_day, so that planned zero-demand days are kept distinct from event-driven demand shifts.

3.1.3 Data Transformation

The transformation phase addresses the principal data quality challenges in the USTore's historical records: inconsistent item naming across billing periods, episodic

rather than continuous date coverage, mid-period catalog changes as new suppliers and SKUs entered the store between 2023 and 2026, and partial price-level grouping in older tally entries where items sharing a price point were not consistently distinguished at the SKU level. Item description standardization will be the first transformation operation applied. Across the available tally sheets, item names exhibit inconsistent formatting, abbreviated labels, and revised product names introduced when supplier catalogs changed. The UST College ID Lace, for example, appears as 'UST College ID Lace,' 'GENERIC Lace,' and 'UST College ID Lace / GENERIC Lace' across different billing months. A controlled vocabulary mapping file will be created prior to transformation, assigning each observed name variant to a canonical item identifier that will serve as the primary key in Dim_Product. All item description strings will be mapped to their canonical identifier during transformation.

To resolve inconsistencies in older records where items sharing a price point were grouped into a single tally row, the transformation pipeline will apply a proportional imputation approach informed by inventory baselines. Specifically, sales volume at a given price point will be distributed across eligible SKUs in proportion to their initial monthly stock levels. These derived item level figures will be tagged with an imputation flag and initially assigned a 0.5 weighting coefficient during FSN classification for downstream model handling, which may be refined during Capstone 2 sensitivity analysis if imputed records are found to significantly skew classification outputs relative to directly captured data. Additionally, the pipeline will incorporate a dynamic event enrichment step. The ETL script will read the user-generated event log and update the corresponding is_event_day flag within the Dim_Date table. This

allows the system to incorporate unscheduled events into the modeling layer, to ensure that the table reflects real-world events. Date field normalization will convert all tally date references to a uniform ISO 8601 format, and the dates extracted during the structural pivot will be validated against the academic calendar to flag any logically implausible dates. Furthermore, for SKUs that first appear in the dataset after the initial 2023 baseline, an entry date field will be assigned in the Dim_Product table. The FSN classification and demand forecasting modules will use this entry date to define each SKU's effective observation window, ensuring that newer items are classified based on their actual active market lifespans rather than a uniform timeframe in which they did not exist.

Additional transformation operations will include the preservation of supplier identity from the source tally sheets. Because each monthly Excel file organizes its tally entries under supplier section headers, the transformation pipeline will extract the supplier name associated with each item and store it as an attribute in Dim_Product. These two supplier-related attributes, supplier name extracted from the tally sheets and the generalized lead time estimate provided separately by USTore management, are stored directly in Dim_Product rather than in a separate supplier dimension table because the available supplier data is limited to only these two fields. Creating a standalone dimension for just two attributes would add query complexity without providing additional analytical depth. Other transformation operations will include duplicate record identification and resolution using composite keys based on date and item, as well as outlier detection for physically implausible quantities using documented maximum daily delivery volumes as upper bounds. The pipeline will also

derive calculated fields, including daily depletion units, cumulative monthly sales, and an estimated days of supply remaining at the current observed depletion rate. All transformation operations will be implemented as a documented Python script using the pandas library, with each step producing an intermediate validation checkpoint to ensure data integrity. Records that cannot be reliably transformed, including tally entries with unresolvable item descriptions, will be flagged and isolated in a separate exception table rather than deleted, preserving the original source record for manual review. The proportion of flagged records relative to the total dataset will be reported in Chapter 4 as a data quality metric.

3.1.4 Data Loading

The loading phase deposits the transformed, validated dataset into the project's Star Schema dimensional data model. The primary target storage is SQLite, chosen for its zero-configuration deployment. However, if concurrent access requirements or Power BI connectivity constraints arise during Capstone 2 implementation, the storage layer may be migrated to structured CSV files organized per dimension table, or to a PostgreSQL. The loading process will operate as an incremental append for prospective data as each day's new tally records are validated and appended to the fact table at the end of each business day and as a one-time bulk load for the transformed historical records at the beginning of the Capstone 2 implementation phase. The complete loaded dataset will serve as the input to all subsequent analytical

processes, including FSN classification, demand forecasting, and Power BI dashboard population.

3.2 Presentation of Star Schema

The Star Schema designed for the USTore Demand Forecasting and Prescriptive Inventory Management Dashboard follows the dimensional modeling principles established by Kimball and Ross (2013), organizing transactional data into a central fact table surrounded by two dimension tables that provide the descriptive context required for analytical querying.



Figure 5. *Star Schema Dimensional Data Model for the USTore Demand Forecasting and Prescriptive Inventory Management Dashboard*

The central table of the schema is Fact_Sales. Each row in Fact_Sales represents a single recorded sales event at the item level and contains: a surrogate primary key (sale_id), two foreign keys linking to the dimension tables (product_id and date_id), the quantity sold (quantity_sold), and two derived fields calculated during ETL transformation (cumulative_monthly_units and daily_depletion_rate). Two additional Boolean flag fields are included: imputation_flag indicates whether the sales

record was derived through proportional allocation from a price grouped historical entry rather than directly captured from an itemized source; tally_date_flag distinguishes historical tally sheet observations, which are episodic point in time entries, from Digital Tallying Interface records, which are continuous daily observations.

These flags allow the forecasting and classification operations to weight records differently based on their source reliability.

The first dimension table is Dim_Product. Each row represents one unique USTore SKU and contains: a surrogate primary key (product_id), the canonical item name (item_name), the product category (category), the assigned price point (unit_price_php), the name of the consignment supplier assigned to the item (supplier_name), a generalized estimated lead time in calendar days (lead_time_days) to support the Reorder Point calculation in Section 3.3.3, the FSN classification assigned after the classification run (fsn_class populated as F, S, or N), the entry date marking the first billing month in which the item appears in the tally records (entry_date), and a record status field (is_active). The supplier_name field is derived from the supplier section headers present in the USTore's existing tally sheet records. The lead_time_days field is a generalized operational estimate relayed by USTore management rather than a value extracted from transactional data. It is initially assigned as a uniform default across all products and can be refined on a per-supplier or per-product basis as more precise lead time information becomes available. These two supplier-related fields are stored directly in Dim_Product rather than in a separate Dim_Supplier table because the available supplier data from the USTore is limited to only these two attributes. Under the Kimball methodology, a dimension with only two

descriptive fields does not justify a standalone table, as the additional join required for every analytical query would add complexity without providing meaningful analytical depth (Kimball & Ross, 2013). This denormalization also simplifies the Power BI dashboard's batch sales report generation, as supplier-level grouping can be performed directly from the product dimension without requiring an additional table relationship. The entry_date field is particularly important for newer SKUs introduced after the initial 2023 baseline, as it defines each item's effective observation window for FSN classification.

The second dimension table is Dim_Date. Each row represents one calendar date within the data scope and contains: a surrogate primary key (date_id), the calendar date in ISO 8601 format (calendar_date), a semester identifier (semester_id), the week number within the academic semester (semester_week), and a set of Boolean event flags populated from the UST academic calendar: is_enrollment_period, is_exam_week, is_event_day, is_sem_break, is_tally_date, and is_store_closed. Three of these fields, is_enrollment_period, is_exam_week and is_sem_break, drive the academic-calendar adjustment of the 30-day demand forecasts (Section 3.3.2). The is_store_closed flag is separate from is_event_day because the two represent opposite demand signals: is_event_day marks dates with unusual demand activity such as university celebrations or campus events, while is_store_closed marks dates with zero expected demand due to planned closures or suspensions. While Dim_Date remains a fixed dimension, is_event_day will be dynamically updated via the Event_Log table and is_store_closed will be dynamically updated via the holiday mapping toggle in the Digital Tallying Interface. This ensures

that both event-driven and closure-driven date classifications are reflected in analytical querying.

A supporting configuration table, Dim_Parameters, is included outside the star schema structure to store system-wide input values used in the EOQ, ROP, and Safety Stock calculations. Each row represents one configurable parameter and contains a parameter name, its current value, the unit of measurement, and the date it was last updated. This table allows USTore management to update cost and lead time inputs directly without requiring changes to the underlying calculation scripts.

A supporting operational table, Event_Log, is included outside the star schema structure to capture demand-influencing events that are not published in the official UST collegiate calendar, such as college-specific celebrations, sports fests, acquaintance parties, and token seasons. Each row in Event_Log represents one logged event and contains a surrogate primary key (event_id), the date of the event (event_date), a descriptive label (event_name), additional context (event_description), the staff member who logged it (created_by), and the date the record was created (date_logged). Event_Log does not connect directly to Fact_Sales and therefore does not participate in analytical queries as a dimension table. Instead, it functions as an operational input to the ETL pipeline, which reads Event_Log during each processing cycle and updates the is_event_day flag in the corresponding Dim_Date rows for the logged event dates. This design allows the event flags to remain current as new unscheduled events are discovered without requiring event-specific Boolean columns for each individual event in Dim_Date, since the general-purpose is_event_day flag is sufficient. Event_Log is populated through the event flagging feature in the Digital

Tallying Interface, where staff can mark any date as an event day with a short descriptive label.

Dim_Inventory table is omitted from the schema, because inventory level is a derived calculation by subtracting the cumulative_monthly_units in Fact_Sales from the beginning monthly stock levels, rather than an independent descriptive entity. Storing it as a dimension would create a rapidly changing dimension that violates the standard Kimball methodology. Beginning monthly stock levels, sourced from the USTore's physical inventory count sheets, are ingested as a reference lookup table outside the Star Schema and joined to Fact_Sales at query time to derive estimated current stock positions.

3.3 Business Analytics Model and Testing

3.3.1 Fast, Slow, and Non-moving (FSN) Analysis Model

The Fast, Slow, and Non-moving Analysis model classifies each active SKU within the USTore's product catalog according to its average sales velocity which is calculated from the dataset, historical tally records, and prospective Digital Tallying Interface records combined. Imputed records receive a 0.5 weighting coefficient to reduce their influence on classification boundaries relative to directly captured data.

The primary velocity metric used for classification is the Average Daily Units Sold (ADUS), calculated for each SKU as the total confirmed units sold divided by the number of active selling days in the observation window. The number of active selling days is the number of distinct days on which the item appears in the records, including recorded zero-sale days and excluding days on which it was out of stock. The tally

sheets from October 2024 onward and the Digital Tallying Interface are both daily records, so this approximates the days on which the item was on sale.

Each SKU's observation window will begin from the item's entry_date field in Dim_Product rather than from a uniform start date of August 2024. The Stock Depletion Rate (SDR) rule was not implemented, because beginning stock is recorded for only a minority of items; it is moved to the Recommendations. Fast-moving items with few selling days are flagged as High-Velocity Limited instead.

Classification thresholds will be determined empirically, with the top 20 percent of the ADUS distribution serving as the initial boundary for the Fast-moving category. An item with no recorded sale in the last 90 days is not classified as Fast-moving, whatever its ADUS; it is classified as Slow-moving. This starting threshold is informed by the Pareto Principle and is consistent with the application of Pareto-based FSN classification boundaries in comparable retail and institutional inventory studies (Tambunan et al., 2018; Mohammed et al., 2024). The 20th percentile cutoff is treated as a data-informed starting point rather than a fixed rule. During the exploratory data analysis phase of Capstone 2, the actual ADUS distribution of the full USTore catalog will be examined for natural break points in the ranked velocity curve, and the final threshold will be adjusted and documented in Chapter 4 if the observed distribution reveals a significantly different boundary. An item classified as Fast-moving with fewer than 30 active selling days is flagged as High-Velocity Limited (is_hvl = 1). The flag marks lower confidence in the classification; the item is still forecast and replenished as Fast-moving.

To assess the robustness of the Pareto-based classification boundary, a threshold sensitivity analysis will be conducted by rerunning the FSN classification at the 75th and 85th percentile cutoffs in addition to the primary 80th percentile threshold. If the majority of SKUs retain their assigned category across all three thresholds, the 80th percentile boundary is confirmed as stable. SKUs that shift between categories across thresholds will be flagged as borderline items and reported in Chapter 4 with a recommendation for manual review by USTore management. The FSN classification output will be stored in the fsn_class field of Dim_Product and will drive category-specific decision rules across the system. For fast-moving items, the system will produce 30-day demand forecasts, calculate Reorder Points and Safety Stock at the designated service level, and compute EOQ to determine the optimal reorder quantity. These items receive the highest analytical priority because they represent the SKUs where stockouts carry the greatest revenue impact. The Strategic Re-run Advisory is not implemented in this phase and is moved to the Recommendations, together with the Stock Depletion Rate rule it depends on. High-Velocity Limited items are reordered through the same Reorder Point as other fast-moving items. For slow-moving items, the system will not produce item-level demand forecasts, as their sales volumes are insufficient for reliable time-series modeling; those with sales in the last 365 days still receive a Reorder Point, Safety Stock and EOQ calculated from their actual sales (Section 3.3.3). In addition, the dashboard will flag slow-moving items with an advisory recommending promotional intervention such as bundled pricing, placement adjustment, or targeted marketing during high-traffic events to increase turnover. Slow-moving items that remain below a configurable velocity threshold for

two consecutive semesters will be escalated to a "Review for Discontinuation" status. For non-moving items, that is items with zero recorded sales within the observation window, the system will exclude them from all replenishment calculations and flag them in the dashboard with a recommendation to evaluate whether the item should be returned to the consignment supplier, marked down for clearance, or formally discontinued from the catalog. The prescribed action for each category will be displayed in the Power BI dashboard's FSN classification view as a decision rule card alongside the item's velocity metrics, so that USTore management can act on the classification output directly.

3.3.2 Demand Forecasting Model

Demand will be forecast at two levels: for each product category, and for each SKU classified as fast-moving, including 'High-Velocity Limited' items, by the FSN Analysis, using daily sales derived from both the transformed historical records and the prospective Digital Tallying Interface data. Category demand is forecast for each of the 12 forecast categories with a six-month (180-day) moving average: the category's average daily sales over the most recent 180 days, multiplied by 30 to give the forecast for the next 30 days. Each fast-moving SKU is forecast with an equal blend of two components: the item's share of its category's sales over the last 30 days, applied to the category forecast, and the Teunter-Syntetos-Babai (TSB) method for intermittent demand (Teunter, Syntetos & Babai, 2011). The TSB smoothing parameters are set at alpha = beta = 0.05. The academic calendar enters both forecasts through a calendar

adjustment rather than as model regressors: using the is_enrollment_period, is_exam_week and is_sem_break flags derived from the Dim_Date table, each 30-day forecast is lowered when the coming 30 days hold a quieter mix of days (such as semestral breaks or examination weeks) than the period it was learned from; it is never raised. Each adjusted 30-day total is then spread across the days by the category's weekday pattern. The semester_week field, the is_store_closed flag and the is_event_day flag are not forecast inputs: semester_week resets each term, closures are often recorded after the fact, and the use of event flags as a forecast input is moved to the Recommendations until enough events have been logged.

The forecast horizon will be set at 30 calendar days from the date of each run. This horizon is selected because it aligns with the USTore's monthly billing and supplier compensation cycle. A 30-day window provides sufficient lead time for restocking decisions given the estimated supplier lead times documented in Section 3.1.1, while avoiding the diminishing accuracy that longer-horizon forecasts would produce given the limited volume of clean historical data currently available. The forecast will be regenerated at the start of each billing month during the Capstone 2 implementation phase. If the system is maintained beyond the capstone period, the rerun frequency may be adjusted to a biweekly or rolling basis as the accumulated dataset grows and forecast stability improves.

The forecasting methods were not fixed in advance: candidate methods, namely Prophet, ARIMA, exponential smoothing, the intermittent-demand methods Croston, SBA and TSB, and moving averages, were compared on the USTore's own sales data under the validation protocol described in Section 3.3.4, and the

comparison is reported in Chapter 4. A category total combines the sales of every item in the category and therefore records sales on more days than any single item, and simple averages of this kind are often difficult to improve upon and serve as a standard baseline in forecasting practice (Hyndman & Athanasopoulos, 2018). TSB is used for items because most item-days in the USTore's records show no sales: it tracks separately how often an item sells and how much it sells when it does, and its forecast decays when an item stops selling rather than holding at the level of the last sale.

The tally sheets from October 2024 onward are daily records; earlier months contain fewer recorded dates, which limits the reliability of any forecasting method on that part of the history. This is a known constraint of the historical dataset. Forecast reliability for periods covered by the earlier, sparser tally records is expected to be lower than for periods covered by continuous daily observations produced by the Digital Tallying Interface. The split-period evaluation described in Section 3.3.4 will quantify this difference directly. Once the Digital Tallying Interface is deployed and begins producing continuous daily observations, those records will be appended to the sales history and the forecasts will be recomputed to incorporate the improved data quality.

Before forecasting each SKU, the system will perform a data sufficiency check on the existing historical dataset. Preliminary inspection of the 2023-2026 tally records confirms that a subset of core apparel items possesses sufficient historical tally dates to support initial modeling, though the exact catalog-wide count of qualifying SKUs will be formalized during the Capstone 2 ETL phase. Each SKU's available time series will be assessed for the minimum number of observations required for stable parameter

estimation. All fast-moving SKUs are forecast with the selected model and validated with the 80/20 walk-forward split described in Section 3.3.4. A fast-moving SKU with fewer than 30 active selling days is flagged as High-Velocity Limited to mark lower confidence in its classification, and it is still forecast and replenished as fast-moving; at present no fast-moving SKU falls under this condition, as all 30 have at least 66 days with sales. The observation counts and sufficiency classification for all SKUs will be reported in Chapter 4 as part of the model validation results.

The forecasts are evaluated under the protocol in Section 3.3.4, against a MAPE target of 20 percent or lower for standard academic semester periods and a naive baseline that repeats the previous period, and the result is reported whether or not the target is met. The forecasts do not drive the Reorder Point, Safety Stock and Economic Order Quantity calculations, which use each item's actual sales (Section 3.3.3). The forecasts are reported for planning and presented in the Power BI dashboard's Demand Forecast View.

3.3.3 Reorder Point, Safety Stock, and Economic Order Quantity Calculation

For each fast-moving and slow-moving SKU with sales in the last 365 days, three prescriptive inventory parameters are computed: the Reorder Point (ROP), the Safety Stock level, and the Economic Order Quantity (EOQ). SKUs with no sales in that period are flagged 'insufficient data - manual review' and receive none. Together, these three values answer the two core questions of inventory replenishment: when to place a restocking request and how many units to order (Waters, 2003).

Reorder Point. The Reorder Point is the stock level at which a replenishment order must be placed to ensure that inventory does not fall to zero before the replacement stock arrives, calculated as follows:

$$\text{ROP} = (\text{Average Daily Demand} \times \text{Lead Time in Days}) + \text{Safety Stock}$$

Average Daily Demand is the SKU's actual average daily sales over the last 365 days; the 30-day forecast is shown alongside but does not enter the ROP. Lead Time in Days is sourced from the lead_time_days field in Dim_Product for the SKU's assigned consignment supplier.

Safety Stock. Safety Stock provides a buffer against forecast uncertainty and supply variability, and is calculated as follows (Aucamp et al., 2020)

Safety Stock is sized from each item's own history: over past test periods the system measures how far demand during one lead time exceeded the item's average rate, and sets the buffer at a quantile of those shortfalls. The quantile depends on the item's service tier: 'servable' items get the smallest quantile (up to 0.95) that meets at least 90% of their demand; 'partial' items stop where extra stock buys little extra service; 'not stockable' items get no safety stock and are flagged as made-to-order candidates. Where an item has too little history to measure its shortfalls, $Z \times \sigma \times \sqrt{\text{Lead Time}}$ is used instead (no item at present); it is also computed and stored for every item for comparison.

Economic Order Quantity. The EOQ determines the optimal number of units to order per replenishment cycle by minimizing the total inventory cost, which is composed of two competing cost components: the ordering cost incurred each time a

restocking request is placed, and the holding cost incurred for each unit stored over time (Harris, 1913; Waters, 2003).

The EOQ is calculated as follows:

$$EOQ = \sqrt{((2 \times D \times S) / H)}$$

In this formula, D is the SKU's actual sales over the last 365 days. S is the ordering cost per replenishment cycle, estimated in consultation with USTore management and the UST Purchasing Office as described in Section 3.1.1. H is the holding cost per unit per year, similarly estimated during the Capstone 2 onboarding phase. The EOQ formula assumes a constant demand rate and a fixed ordering cost, both of which are simplifications of the USTore's actual operating environment. However, the EOQ output serves as a data-informed starting point for order quantity decisions that is objectively better than the experiential guesswork currently used by the store. To validate whether an EOQ-derived reorder quantity is truly optimal, the system will compute the total inventory cost (ordering cost + holding cost) at the EOQ value and compare it against the total cost at alternative order quantities such as 0.5x EOQ and 2x EOQ. If the EOQ value consistently produces the lowest total cost across the sensitivity range, it is confirmed as the optimal quantity for that SKU. This sensitivity check will be reported per SKU in Chapter 4. Because the cost inputs are provisional estimates, EOQ exceeds a full year of demand for 206 to 210 of the 210 SKUs; EOQ is reported as calculated, and the dashboard also suggests an order-up-to quantity (the Reorder Point plus 30 days of demand, minus stock on hand). All computed values, including ROP, Safety Stock, and EOQ, will be expressed in units of

physical stock count and stored as calculated fields in a summary table refreshed at each model rerun. The computation will be implemented in Python and will feed directly into the Power BI dashboard's prescriptive restocking advisory system.

3.3.4 Model Validation and Testing Protocol

The forecasting models will be validated using walk-forward validation within an 80/20 split: the first 80 percent of the time series (2 May 2024 to 16 February 2026) is used for training, and the remaining 20 percent (17 February to 31 July 2026) is forecast in 30-day steps, with each model refitted on all data up to each step (Brownlee, 2020). Prophet and the naive baseline are validated the same way for comparison. The candidate methods were compared on the last 12 months of data, which overlap the test period. The model will generate forecasts for the test period, and forecast accuracy will be measured using three metrics: Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). The primary acceptance criterion is a MAPE of 20 percent or lower on the test set during standard academic semester intervals, supplemented by RMSE and MAE reported as diagnostic metrics per SKU across all periods, including semestral breaks; no separate MAE threshold is set for break periods. The reorder recommendations, which the forecasts do not drive (Section 3.3.3), are judged separately against the USTore acceptance standard, four conditions set before measurement: (1) every eligible item gets a reorder point or an explicit flag, and the items priced in advance account for at least 90% of the demand that follows; (2) the reorder points meet more

demand than stocking what sold in the last lead time, at every test period; (3) no simpler setting meets as much demand with less stock; (4) every result states how much of its test period is observed data. To contextualize these results, a naive baseline forecast will be established prior to model evaluation, in which the predicted demand for each period equals the observed demand from the immediately preceding period. Each model's MAPE, MAE, and RMSE will be compared against the naive baseline. Models that fall short are retuned and compared with alternative methods. If no forecast measurably beats the naive baseline, the forecasts are reported for planning only and the restocking calculations use actual sales (Section 3.3.3). MAE is reported for each fast-moving SKU in Chapter 4 as a diagnostic measure; no item-level MAE threshold is set.

No SKU falls under the condition for leave-one-out cross-validation (fewer than 60 total observations): all 30 fast-moving SKUs have at least 66 days with sales, so the same 80/20 walk-forward test applies to every SKU and leave-one-out cross-validation is not needed.

The split-period evaluation, which measures forecast performance separately for periods covered predominantly by historical tally data and periods covered by Digital Tallying Interface data, is not yet applicable: the comparison starts once Digital Tallying Interface records are available in sufficient volume, and none exist yet (all 84,430 sales rows come from the tally sheets). Meanwhile each result states how much of its test period is observed data (condition 4 of the USTore acceptance standard). Once applicable, the split-period evaluation will provide a direct empirical basis for quantifying the improvement in forecast accuracy attributable to clean digital

data collection, which represents one of the core value propositions of the proposed system.

Given that MAPE becomes unreliable when actual sales values approach zero, as occurs during semestral break periods, MAE is reported as the main error measure for low-volume intervals; it is not used as a pass/fail threshold. MAPE, MAE and RMSE are reported separately for semester and break periods for each fast-moving SKU. The split-period evaluation will separately report MAPE, MAE, and RMSE for high-volume academic periods and low-volume break periods, providing a complete picture of model reliability across the full range of USTore's demand environment.

3.4 Business Analytics Tools, Techniques, and Specific Applications

The system will require a structured, multi-layered analytical pipeline to transition from the USTore's fragmented raw data into actionable restocking insights. Table 3 describes the specific business analytics tools, techniques, and specific applications for each functional layer of the system architecture. The stack included in the table was strategically chosen and assembled to address the USTore's operational constraints, errors, and historical data irregularities. The framework in the table outlines a logical progression, starting with the data capture and ETL processing of the historical data into a star schema in SQLite. Afterwards, the data is subjected to FSN analysis and time-series forecasting. Finally, a model that calculates ROP and safety stock levels which are then displayed in the final dashboard.

The proposed system integrates three progressive levels of business analytics. At the descriptive analytics level, the FSN classification module and Power BI stock status views summarize historical and current inventory behavior, answering the operational question of what has occurred within the USTore's sales cycle. At the predictive analytics level, the demand forecasting module, a six-month moving average per category and a blend of category share and TSB per fast-moving item adjusted for the UST academic calendar, estimates future demand, answering the question of what demand levels are expected in forthcoming periods. At the prescriptive analytics level, the Reorder Point, Safety Stock, and Economic Order Quantity calculation module converts each item's actual sales over the last 365 days into specific, actionable restocking decisions by solving for the order quantity that minimizes total inventory cost while maintaining the service level set by each item's service tier. The dashboard's calendar-contextual interpretation cards surface forward-looking narrative guidance tied to upcoming demand-influencing events. Together, these components answer the operational question of what management should do, when they should do it, and how much they should order.

This three-tier analytics architecture ensures that the system delivers value not only as a reporting tool but as a genuine decision-support instrument aligned with the operational reality of the USTore.

Table 3. Business Analytics Tools, Techniques, and Specific Applications for the Proposed System.

Layer	Tool	Technique	Analytics	Specific Application
Data Capture	Flask or React + SQLite / Google Sheets	Structured form-based data entry with field validation	**Descriptive**	Digital replacement for paper tally sheets; BIR-compliant non-transactional logging of daily USTore sales by item, quantity, date.
Data Processing	Python 3.x with pandas library	ETL with wide-to-long pivot; proportional allocation for price-grouped records; tally-date normalization	**Descriptive**	Ingestion, cleansing, and standardization of USTore tally and inventory count records; derivation of per-date depletion figures
Data Modeling	SQLite database with Star	Dimensional data modeling	**Descriptive**	Structured storage of Fact_Sales, Dim_Product, and Dim_Date tables

	Schema architecture	(Kimball methodology)		enabling efficient analytical querying
Inventory Classification	Python 3.x with pandas library	FSN Analysis using Pareto-based empirical thresholds; variable observation window per SKU	**Descriptive**	Segmentation of all active USTore SKUs into F, S, and N velocity categories to prioritize forecasting resources and dashboard alerts
Demand Forecasting	Python 3.x (pandas, NumPy)	Six-month moving average per category; category share and TSB blend per fast-moving item;	**Predictive**	30-day demand forecasts per category and per fast-moving SKU

		academic-calendar adjustment; weekday day pattern		
Replenishment Calculation	Python 3.x (custom calculation script)	Reorder Point and Safety Stock formulae with service levels set by each item's service tier; Economic Order Quantity (EOQ) optimization	**Prescriptive**	Unit-level restocking thresholds and optimal order quantities per fast-moving and slow-moving SKU using 365-day actual demand and a safety stock sized from measured shortfalls, documented lead times, and estimated ordering and holding costs
Visualization and Reporting	Microsoft Power BI Desktop (free tier)	Interactive dashboard with KPI cards, trend charts,	**Descriptive, Predictive, Prescriptive**	Stock status monitoring, FSN view, demand forecast visualization, and automated PDF export of

		and color-coded alert indicators		monthly batch sales reports for UST Purchasing Office and Finance Department

The layered architecture described in Table 3 reflects a deliberate design philosophy of deploying open-source, low-infrastructure tools at every stage of the analytics pipeline. The approach ensures that the completed system is maintainable by the USTore without ongoing licensing costs, operable without enterprise information technology support, and extensible as the volume of clean digitally captured data grows over successive academic semesters. The Python analytics layer is designed as a modular pipeline where individual components, the ETL script, the FSN classification module, the demand forecasting module, and the ROP calculation module can each be independently updated or replaced as the project's data quality and analytical requirements evolve.

The foundational layers of the proposed architecture establish a robust, compliant data pipeline, beginning with Data Capture. Utilizing lightweight web frameworks (Flask or React) combined with SQLite or Google Sheets, this layer serves as a BIR-compliant, non-transactional digital replacement for paper tally sheets, accurately logging daily item depletion without processing monetary transactions. Following capture, the Data Processing layer leverages Python 3.x and the *pandas* library to execute a comprehensive ETL pipeline. This stage cleanses and

standardizes the raw records through wide-to-long table pivoting, tally-date normalization, and proportional allocation for older price-grouped entries. Once processed, the Data Modeling layer ingests these derived depletion figures into an SQLite database structured around a Star Schema using the Kimball methodology. This organizes the operational data into a central *Fact_Sales* table surrounded by *Dim_Product*, and *Dim_Date.*

Rather than isolating these tools, the proposed architecture functions as a continuous, integrated pipeline. The system sequentially captures and standardizes historical and live data into a robust dimensional model, establishing a reliable descriptive baseline. From there, the analytical engine applies velocity-based item classification and academic-calendar-adjusted time-series forecasting to generate predictive demand insights. Ultimately, the pipeline converts each item's actual sales over the last 365 days into prescriptive outputs, specifically Reorder Points and Safety Stock alerts surfaced through an automated, interactive dashboard to directly support USTore management and Finance Department workflows.

3.5 Risk Assessment and Mitigation

The implementation of the proposed system is subject to risks across the data pipeline, forecasting model, and stakeholder dimensions of the project. Table 4 identifies nine primary risks, assesses their likelihood and impact, and specifies the mitigation measure integrated into the system design or implementation protocol for each (Project Management Institute, 2021).

Table 4. Project Risk Register for the USTore Demand Forecasting and Inventory Management Dashboard

Risk	Category	Likelihood	Impact	Mitigation
Historical tally records contain unresolvable item description ambiguities that prevent reliable ETL transformation	Data Quality	High	High	Controlled vocabulary mapping file applied during transformation; unresolvable records isolated in an exception table and reported as a data quality metric in Chapter 4
The forecasting model fails to demonstrate meaningful improvement over the naive baseline	Model Performance	Moderate	High	Candidate methods, including Prophet, compared with the naive baseline on the same 80/20 test; if none improves on it, the forecasts are reported for planning only, and the reorder recommendations are checked separately against stocking

Risk	Category	Likelihood	Impact	Mitigation
				without a model (acceptance standard, Section 3.3.4).
Low transaction volumes during semestral breaks produce data-sparse intervals that distort forecasting model parameters	Data Sparsity	High	Moderate	Semestral breaks and examination weeks handled by the academic-calendar adjustment, which lowers the 30-day forecast when a quieter mix of days is ahead; MAE reported as the main error measure for break periods (Section 3.3.4)
USTore staff revert to paper-based tallying after Digital Tallying Interface deployment	Adoption	Moderate	High	Interface designed for minimum interaction steps with field-level validation; data completeness rate monitored and reported to USTore management monthly

Risk	Category	Likelihood	Impact	Mitigation
USTore management delays or restricts access to historical tally records during data gathering	Stakeholder Cooperation	Low	High	Formal data-sharing agreement established at project onboarding; fallback scope defined for reduced historical coverage
Mid project changes to the merchandise catalog introduce new or discontinued SKUs.	SKU Catalog Volatility	Moderate	Moderate	entry_date and is_active fields in Dim_Product track new SKUs from their effective introduction date and exclude discontinued items from active forecasting without disrupting historical records
Non-sale inventory removals such as returns, damages, or internal transfers are	Data Integrity	Moderate	High	A transaction_type field is included in the Digital Tallying Interface schema to distinguish sale events from non-sale

Risk	Category	Likelihood	Impact	Mitigation
misrecorded as sales transactions, inflating demand estimates and distorting FSN classification				removals at the point of data entry; ETL pipeline filters non-sale records from the demand calculation input while retaining them in the data warehouse for inventory reconciliation purposes
FSN classification boundaries misassign SKUs near velocity thresholds, causing borderline items to receive inappropriate forecasting treatment	Classification Accuracy	Moderate	Moderate	Threshold sensitivity analysis conducted during model validation by testing alternative percentile cutoff values and comparing the resulting classification outputs; SKUs that shift categories under minor threshold adjustments are flagged for manual review by USTore management before finalizing the classification output

Risk	Category	Likelihood	Impact	Mitigation
Staff encoding lapses misrecorded as zero sales.	Data Integrity	Moderate	High	Implementation of the feature of an intra-day completeness flag and mid-day checkpoint alerts through the Digital Tallying Interface and dashboard.

The two risks with the highest combined likelihood and impact are data quality degradation and Digital Tallying Interface non-adoption. Both are addressed at the layer of the system where they originate: the ETL pipeline handles ambiguous historical records through a controlled vocabulary mapping step and isolates unresolvable entries into an exception table rather than discarding them, while the interface is designed for minimum interaction steps and monthly completeness monitoring to sustain consistent usage. Data sparsity during semestral breaks and SKU catalog volatility are structural features of the USTore's operating environment rather than unpredictable events, and both are accounted for directly in the data model through the Dim_Date calendar flags that drive the academic-calendar adjustment of the forecasts and the entry_date and is_active fields in Dim_Product respectively. Non-sale inventory removals being misrecorded as sales is mitigated at the data entry layer through a transaction_type field that distinguishes sale events from returns,

damages, and internal transfers before the data reaches the analytics pipeline. FSN classification boundary misalignment is addressed through threshold sensitivity analysis during model validation, where alternative percentile cutoffs are tested and borderline SKUs are flagged for manual review. Staff encoding lapses, where sales occur but are not recorded, are mitigated through the intra-day completeness audit and the Unverified Zero flag, which prevent unconfirmed zero-sales entries from being treated as true demand observations in the forecasting model. The remaining risks, the forecasting models failing to outperform the naive baseline across the multi-metric evaluation framework and restricted access to historical records, are lower in likelihood but high in impact, and each has a defined fallback: if no forecast improves on the naive baseline, the forecasts are reported for planning only and the reorder recommendations are checked separately against stocking without a model; and a reduced-coverage scope is agreed upon with USTore management before data gathering begins.

Bibliography

Adamson, C. (2006). Star schema basics. In Mastering data warehouse aggregates: Solutions for star schema performance. Wiley. https://www.oreilly.com/library/view/mastering-data-warehouse/9780471777090/9780471777090_star_schema_basics.html

Athique, A. (2025). Abot kamay: Embedding digital transactions in the Philippines. International Journal of Cultural Studies, 29(1), 120–137. https://doi.org/10.1177/13678779251348945

Axsäter, S. (2006). Inventory control (2nd ed.). Springer.

Bouchard, S., Gamache, S., & Abdulnour, G. (2023). Strategy using modularity tools to operationalize mass customization in manufacturing small and

medium-sized enterprises. Cleaner Logistics and Supply Chain, 9, 100123. https://doi.org/10.1016/j.clscn.2023.100123

Boyd, J., Porter, J., Ferguson, M., Graham, J., Brooks, E., Hayes, D., Peterson, S., & Oloyede, J. (2025). Manual record keeping as a driver of leakage. ResearchGate. https://www.researchgate.net/publication/396248887_Manual_record_keeping_as_a_driver_of_leakage

Bradley, A. R., Barbosa, A., Younk, L., Rocha, N., & Nichol, P. F. (2026). The limits of humans in data gathering: Documentation error rates in the electronic health record in the operating room. PubMed Central. https://pmc.ncbi.nlm.nih.gov/articles/PMC12883500/

Brownlee, J. (2020, August 26). Time series forecasting with Prophet in Python. Machine Learning Mastery. https://machinelearningmastery.com/time-series-forecasting-with-prophet-in-python/

Chowdhury, A. R., Paul, R., & Rozony, F. Z. (2025). A systematic review of demand forecasting models for retail e-commerce: Enhancing accuracy in inventory and delivery planning. International Journal of Scientific Interdisciplinary Research, 6(1), 1–27. https://doi.org/10.63125/mbbfw637

Croston, J. D. (1972). Forecasting and stock control for intermittent demands. Operational Research Quarterly, 23(3), 289–303. https://doi.org/10.2307/3007885

Davis, A., Chen, A., Chen, M., & Davis, J. (2023). Viability of mobile forms for population health surveys in low resource areas (arXiv:2310.07888). arXiv. https://doi.org/10.48550/arXiv.2310.07888

Devi, T. P., Bharti, S., & Hassan, R. (2025). Enhancing sales performance analysis through advanced data visualization techniques: Evidence from e-commerce transaction data. International Journal of Scientific Research in Engineering and Management, 9(11). https://doi.org/10.55041/IJSREM53646

Dunford, R., Su, Q., Tamang, E., & Wintour, A. (2014). The Pareto principle. PEARL. https://doi.org/10.24382/swfr-wr17

Famoti, O., Ezechi, O., Ewim, C., Eloho, O., Muyiwa-Ajayi, T., Igwe, A., & Ihechere, A. (2025). *Operational efficiency in retail: Using data analytics to optimize inventory and supply chain management*. International Journal of Scientific Research in Computer Science, Engineering and Information Technology, 11, 1483–1494. https://doi.org/10.32628/CSEIT251112173

Garudasu, S., Balasubramaniam, V. S., Kumar, P., Singh, N., Goel, P., & Goel, O. (2022). Leveraging Power BI and Tableau for advanced data visualization and business insights. International Journal of General Engineering and Technology, 11(2), 153–174.

Gerchak, Y., & Khmelnitsky, E. (2003). A consignment system where suppliers cannot verify retailer's sales reports. International Journal of Production Economics, 83(1), 37–43. https://doi.org/10.1016/S0925-5273(02)00264-5

Han, J., Pei, J., & Tong, H. (2022). Data mining: Concepts and techniques (4th ed.). Morgan Kaufmann.

Hickins, M. (2023). What is retail inventory management? Guide & tips. Oracle. https://www.oracle.com/asean/retail/fashion/retail-inventory-management/

Hua, J. (2023). Challenges and strategies for inventory management in small and medium-sized cosmetic enterprises: A review. *International Journal of Information Technology and Computer Science Applications, 1*(2). https://doi.org/10.58776/ijitcsa.v1i2.30

Hewamalage, H., Ackermann, K., & Bergmeir, C. (2023). Forecast evaluation for data scientists: Common pitfalls and best practices. Data Mining and Knowledge Discovery, 37(2), 788–832. https://doi.org/10.1007/s10618-022-00894-5

Hyndman, R. J., & Athanasopoulos, G. (2018). Forecasting: Principles and practice (2nd ed.). OTexts.

Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. International Journal of Forecasting, 22(4), 679–688. https://doi.org/10.1016/j.ijforecast.2006.03.001

Islam, S. M., & Ferrer, M. A. (2025). Unlocking the cashless shift: Retailers' adoption of digital payment systems in emerging markets. Journal of

Theoretical and Applied Electronic Commerce Research, 20(4), 359. https://www.mdpi.com/0718-1876/20/4/359

Jacob. (2024). Adoption of cashless payments by retail enterprises in Legazpi City. *Philippine E-Journals*. https://ejournals.ph/article.php?id=24282

Jenkins, A. (2026). The role of ERP in supply chain management. NetSuite. https://www.netsuite.com/portal/resource/articles/erp/supply-chain-management-erp.shtml

Joseph, O. (2025). Enhancing business intelligence with data visualization tools. ResearchGate. https://www.researchgate.net/publication/388780215_Enhancing_Business_Intelligence_with_Data_Visualization_Tools

Kimball, R., & Ross, M. (2002). The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling (3rd ed.). Wiley Computer Publishing.

Kolambe, N. (2024). Forecasting the future: A comprehensive review of time series prediction techniques. Journal of Mathematical & Computer Applications, 3(2), 1–7. https://doi.org/10.47363/JMCA/2024(3)180

Kumar, P., Jena, D., Trivedi, S. K., Padhy, R., & Roy, A. D. (2024). Crack spread forecasting for supply chain optimization: A hybrid model using time series and deep learning with Bayesian optimizations. Academy of Marketing Studies Journal, 28(3). https://www.abacademies.org/articles/crack-spread-forecasting-for-supply-chain-optimization--a-hybrid-model-using-time-series-and-deep-learning-with-bayesian.pdf

Laudon, K. C., & Laudon, J. P. (2022). Management information systems: Managing the digital firm (17th ed.). Pearson.

Lightspeed. (n.d.). Retail POS system. https://www.lightspeedhq.com/pos/retail/

Liu, Y., Kalaitzi, D., Wang, M., & Papanagnou, C. (2025). A machine learning approach to inventory stockout prediction. Journal of Digital Economy. https://doi.org/10.1016/j.jdec.2025.06.002

Malhotra, M. K., Mackelprang, A. W., & Jayaram, J. (2017). How inventory consignment programs can improve supply chain performance: A process oriented perspective. Production, 27, Article e20170064. https://doi.org/10.1590/0103-6513.006417

Mia, I. B., Jimenez, S., Habaradas, R., Ranieses, J. J., Javier, C., & Enriquez, J. (2024). Digital technology adoption among Philippine micro-, small-, and medium-sized enterprises: Barriers, enablers & challenges during COVID-19. *Journal of Business, Ethics and Society, 4*(1). https://doi.org/10.61781/4-1I2024/1bmlm

Mohammed, H. O., Abdullahi, S., & Abba, A. (2024). Inventory management methods, approaches, benefits, and challenges. International Journal of Social Sciences and Management Review, 7(4), 164–177. https://doi.org/10.37602/IJSSMR.2024.7410

Morfonios, K., & Ioannidis, Y. (2009). Star schema. In L. Liu & M. T. Özsu (Eds.), Encyclopedia of database systems. Springer. https://doi.org/10.1007/978-0-387-39940-9_888

Mustapha, Y., & Adnan, R. M. (2025). Forecasting retail sales using machine learning models. Journal of Data Science and Intelligent Systems, 6(1), 1–15. https://doi.org/10.47260/jdsis/v6i1.1

Nirmani, I. A. P. (2025). Barriers to digital participation in developing countries: Identifying technological, social, and cultural obstacles to community involvement. GSC Advanced Research and Reviews, 23(2), 61–71.

Onyeanuforo, U. L., & Mohammed, B. B. (2024). Data-driven decision-making in global supply chain management: Transforming operations through analytics and intelligence. Nigerian Journal of Sustainability Research, 2(2), 1–23. https://doi.org/10.5281/zenodo.16949528

Osio, E. R. B., Trinidad, F. L., & Alejandro, G. J. D. (2026). Digital resilience and technology adoption among retail SMEs in post-pandemic Metro Manila. Journal of Economics, Finance and Management Studies, 9(1), 217–231. https://doi.org/10.47191/jefms/v9-i1-21

Park, M.-J., Turner, O., & Becker, T. (2025). Inventory optimization in retail supply chains using deep reinforcement learning. Advances in Management

and Intelligent Technologies.
https://ojs.apspublisher.com/index.php/amit/article/view/470

Prabu, V. P. (2023). AI-powered predictive analytics for retail demand forecasting. *International Journal of Innovative Research in Engineering & Multidisciplinary Physical Sciences, 11*(2).
https://doi.org/10.37082/IJIRMPS.v11.i2.232381

Rahman, M. A., Alam, M. S., & Mrida, M. S. H. (2025). How interactive dashboards improve managerial decision-making in operations management. American Journal of Advanced Technology and Engineering Solutions.
https://ajates-scholarly.com/index.php/ajates/article/view/6

Reason, J. (2000). Human error: Models and management. BMJ, 320(7237), 768–770. https://pmc.ncbi.nlm.nih.gov/articles/PMC1117770/

Röding, T., Steinmann, S., Wagner, G., & Schramm-Klein, H. (2023). A classification of information-oriented PoS technology from customer perception. Journal of Retailing and Consumer Services, 70, Article 103126.
https://doi.org/10.1016/j.jretconser.2022.103126

Saeed, N., Nguyen, S., Cullinane, K., Gekara, V., & Chhetri, P. (2023). Forecasting container freight rates using the Prophet forecasting method. Transport Policy, 133, 86–107. https://doi.org/10.1016/j.tranpol.2023.01.012

Seyedan, M., & Mafakheri, F. (2023). Predictive big data analytics for supply chain demand forecasting: Methods, applications, and research opportunities. Journal of Big Data, 10(1), Article 87.
https://doi.org/10.1186/s40537-023-00769-y

Shaik, A., Khan, I., Dandu, M. M. K., Goel, P., Jain, A., & Shrivastav, E. A. (2024). The Role of Power BI in Transforming Business Decision-Making: A case study on healthcare reporting. Journal of Quantum Science and Technology., 1(3). https://doi.org/10.63345/jqst.v1i3.117

Shapiro, S., & Panvelwala, B. (2026). Prophet as a reproducible forecasting framework: A methodological guide for business and financial analytics. arXiv.
https://doi.org/10.48550/arXiv.2601.05929

Silaen, B. R., Nasution, M., & Muti'ah, R. (2024). Implementation of the ABC analysis to the inventory management. International Journal of Science, Technology & Management, 5(4), 816–825. https://doi.org/10.46729/ijstm.v5i4.1144

Singh, C. (2025). Optimizing Supply Chain Operations with Predictive Analytics. 10.13140/RG.2.2.15758.68169.

Singh, S. (2025). Comparative Analysis of Python's Role in AI and Machine Learning. Laudon, K. C., & Laudon, J. P. (2022). Management information systems: Managing the digital firm (17th ed.). Pearson. https://www.pearson.com/en-us/subject-catalog/p/management-information-systems-managing-the-digital-firm/P200000007007/9780136971276

Srinivasan, E., & Gowsalya, M. (2025). Optimizing data analysis for future sales forecasting using artificial intelligence. International Journal of Novel Research and Development, 10(5), b284–b293. https://www.ijnrd.org/papers/IJNRD2505126.pdf

Syntetos, A. A., & Boylan, J. E. (2005). The accuracy of intermittent demand estimates. International Journal of Forecasting, 21(2), 303–314. https://doi.org/10.1016/j.ijforecast.2004.10.001

Syntetos, A. A., Boylan, J. E., & Croston, J. D. (2005). On the categorization of demand patterns. Journal of the Operational Research Society, 56(5), 495–503. https://doi.org/10.1057/palgrave.jors.2601841

Szabłowski, B. (2026). One global model, many behaviors: Stockout-aware feature engineering and dynamic scaling for multi-horizon retail demand forecasting with a cost-aware ordering policy (arXiv:2601.18919). arXiv. https://doi.org/10.48550/arXiv.2601.18919

Szostek, K., Mazur, D., Drałus, G., & Kusznier, J. (2024). Analysis of the effectiveness of ARIMA, SARIMA, and SVR models in time series forecasting: A case study of wind farm energy production. Energies, 17(19), Article 4803. https://doi.org/10.3390/en17194803

Tambunan, M. M., Syahputri, K., Rizkya, I., Sari, R. M., & Cahyo, M. D. (2018). Storage design using fast moving, slow moving and non moving (FSN)

analysis. MATEC Web of Conferences, 197, Article 14005. https://doi.org/10.1051/matecconf/201819714005

Tate, A., & Smallwood, C. (2021). Comparing the efficiency of paper-based and electronic data capture during face-to-face interviews. PLOS ONE, 16(3), Article e0247570. https://doi.org/10.1371/journal.pone.0247570

Teunter, R. H., Syntetos, A. A., & Babai, M. Z. (2011). Intermittent demand: Linking forecasting to inventory obsolescence. European Journal of Operational Research, 214(3), 606–615. https://doi.org/10.1016/j.ejor.2011.05.018

Oguis, G.F. et al. 2022. Optimization Model of Supply Chain Network Design for Small-scale Banana Flour Production Using Deterministic and Metaheuristic Approach. Philipp J Sci 151(5): 1995–2009. https://doi.org/10.56899/151.05.36

Yerra, S. (2025). Enhancing inventory management through real-time Power BI dashboards and KPI tracking. International Journal of Scientific Research in Computer Science, Engineering and Information Technology, 11(2), 944–951.

Zamil, M. H., Mohiuddin, M., & Mamun, M. N. H. (2024). Business intelligence systems in finance and accounting: A review of real-time dashboarding using Power BI & Tableau. American Journal of Scholarly Research and Innovation, 3(2), 52–79. https://doi.org/10.63125/fy4w7w04